Galois Theory / MM7043 2026W

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1 Fields

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Throughout the semester, we will employ the following conventions:

1. Unless otherwise specified, all rings will be assumed to be unital and commutative.
2. A *ring homomorphism* is a map of rings $\varphi : R \to S$, such that
   i. $\varphi(x + y) = \varphi(x) + \varphi(y)$, for all $x, y \in R$.
   ii. $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$, for all $x, y \in R$.
   iii. $\varphi(1_R) = 1_S$

Definition 1.0.1. Let R be a ring. An element $x \in R$ is called a *unit* or *invertible*, if there exists an element $y \in R$ such that $x \cdot y = 1$.

The set of all units of R form a group under multiplication and will be denoted R^*.

Example 1.0.2.

1. $\mathbb{Z}^* = \{-1, 1\}$
2. $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$
3. For $n \in \mathbb{N}$, it holds that $\mathbb{Z}_n^* = \{m \in \mathbb{Z}_n : \gcd(m, n) = 1\}$.

Definition 1.0.3. A *field* is a ring K such that $K^* = K \setminus \{0\}$.

Example 1.0.4. We recall the following fields:

1. $\mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$
2. $\mathbb{Z}_p$, where $p \in \mathbb{P}$.

Lemma 1.0.5. *A ring R is a field if and only if the only ideals of R are $\{0\}$ and R.*

Proof. Let $I \subseteq R$ be an ideal. If I contains a unit then $I = R$.

"$\Longrightarrow$": Assume R is a field. If $I \subseteq R$ is a non-zero, non-trivial ideal, then I contains a unit, which implies $I = R$.

"$\Longleftarrow$": If R is not a field, then there exists some non-zero, non-unit element $x \in R$. But then the ideal Rx satisfies $\{0\} \subsetneq Rx \subsetneq R$. ∎

Corollary 1.0.6. *A homomorphism of fields is always injective.*

Proof. Let $\varphi : \mathbb{K} \to \mathbb{L}$ be a homomorphism of fields. Then $\ker \varphi$ is an ideal in K. We have $1_{\mathbb{K}} \notin \ker \varphi$, therefore $\ker \varphi = \{0\}$ and φ is injective. ∎

1.1 Field extensions

When constructing new fields, there are two main approaches:

1. The approach "from above", where we consider intersections of fields that satisfy our requirements of an extension, which we will discuss in the following.
2. The approach "from below", where we explicitly construct a new field based on roots of polynomials. We will discuss this approach in the next section.

Lemma 1.1.1. *Suppose $\mathbb{K}$ is a field and $\{\mathbb{F}_\alpha : \alpha \in I\}$ is a family of subfields of $\mathbb{K}$. Then*

$$\bigcap_{\alpha \in I} \mathbb{F}_\alpha$$

is a subfield of $\mathbb{K}$.

Definition 1.1.2. Suppose $\mathbb{L}_0 \subseteq \mathbb{L}$ are fields and $\alpha_1, \alpha_2, ... \in \mathbb{L}$. Then we define

$$\mathbb{L}_0(\alpha_1, \alpha_2, ...)$$

as the intersection of all subfields of $\mathbb{L}$ that contain $\mathbb{L}_0$ and $\alpha_1, \alpha_2, ...$, and will be called the *extension* of $\mathbb{L}_0$ generated by $\alpha_1, \alpha_2, ...$

It is clear that such an extension is the smallest subfield of $\mathbb{L}$ that contains $\mathbb{L}_0$ and the elements $\alpha_1, \alpha_2, ...$

Example 1.1.3.

1. Consider the fields $\mathbb{Q} \subseteq \mathbb{R}$, then one can show
$$\mathbb{Q}\big(\sqrt{2}\big) = \big\{p + q\sqrt{2} : p, q \in \mathbb{Q}\big\}.$$
2. Since the element $1_{\mathbb{Q}}$ generates $\mathbb{Q}$ as a field, it follows that $\mathbb{Q}$ has no proper subfields, i.e. $\mathbb{Q}$ is a *primary field*.
3. The field $\mathbb{F}_p$ is a primary field for any $p \in \mathbb{P}$.

Definition 1.1.4. Let $\mathbb{F}$ be a field. The *primary subfield* of $\mathbb{F}$ is the subfield generated by the element $1_{\mathbb{F}}$.

Definition 1.1.5. Let R be a ring, then its *characteristic* is defined by

$$\text{char } R := \begin{cases} \min\Big\{n \in N : \sum_{j=1}^{n} 1_R = 0\Big\}, & \text{if such an } n \text{ exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 1.1.6. *Let $\mathbb{F}$ be a field.*

1. *If* $\text{char } \mathbb{F} = 0$*, then the primary subfield of* $\mathbb{F}$ *is*[1] $\mathbb{Q}$*.*
2. *If* $\text{char } \mathbb{F} = p \in \mathbb{P}$*, then the primary subfield of* $\mathbb{F}$ *is* $\mathbb{F}_p$*.*

Lemma 1.1.7. *Suppose $M \subsetneq \mathbb{R}$ is a maximal ideal. Then the quotient R/M is a field.*

Remark 1.1.8. Suppose $\mathbb{K}$ is a field and consider $\mathbb{K}[x]$, the ring of polynomials in x with coefficients in $\mathbb{K}$. Recall that $\mathbb{K}[x]$ is a principle ideal domain. A non-trivial ideal $(f(x))$ is maximal in $\mathbb{K}[x]$ if and only if $f(x)$ is an irreducible polynomial. In particular, $\mathbb{K}[x]/(f(x))$ is a field whenever $f(x)$ is irreducible over $\mathbb{K}$.

Moreover, $\mathbb{K}[x]/(f(x))$ contains $\mathbb{K}$ as a subfield, via the homomorphisms

[1] We always mean "up to isomorphism".

$$\mathbb{K} \hookrightarrow \mathbb{K}[x] \to \mathbb{K}[x]/(f(x)).$$

Therefore, for every irreducible $f(x)$ the field $\mathbb{K}[x]/(f(x))$ is a field extension of $\mathbb{K}$.

Note that $f(x)$ being irreducible with $\deg f > 1$ implies that $f(x)$ has no roots in $\mathbb{K}$. But $f(x)$ always has a root in $\mathbb{K}[x]/(f(x))$, because $f(x)$ is zero as an element of $\mathbb{K}[x]/(f(x))$.

The above demonstrates, roughly, that there are relationships:

$$\begin{array}{rcl} \text{Polynomials} & \longleftrightarrow & \text{Field extensions} \\ \text{Properties of roots} & \longleftrightarrow & \text{Symmetries of field extensions} \end{array}$$

Practically, we want some criteria to determine if a polynomial is irreducible.

Remark 1.1.9. If $\mathbb{K}$ is a field and $p(x) \in \mathbb{K}[x]$, $\deg p > 1$ is a polynomial with a root in $\mathbb{K}$, then $p(x)$ is not irreducible. Indeed, for any $a \in \mathbb{K}$ one can write

$$p(x) = q(x)(x - a) + p(a)$$

for some polynomial $q(x) \in \mathbb{K}[x]$. If a is a root of $p(x)$, this demonstrates that $p(x)$ is not irreducible.

Note that the converse fails: The polynomial $\left(x^2 + 1\right)^2$ has no roots in $\mathbb{R}$, yet is not irreducible.

Lemma 1.1.10. *For a field $\mathbb{K}$, a polynomial $p(x) \in \mathbb{K}[x]$ with $2 \leq \deg p \leq 3$ is irreducible if and only if it has no roots in $\mathbb{K}$.*

Lemma 1.1.11 (Gauss)**.** *Suppose $p(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{Z}[x]$ is* primitive*, that is $\gcd(a_0, a_1, ..., a_n) = 1$. Then $p(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.*

Lemma 1.1.12 (Eisenstein's Criterion)**.** *Let $q(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{Z}[x]$ and suppose there exists a prime $p \in \mathbb{P}$ such that:*

1. *$p \nmid a_n$*
2. *$p \mid a_0, ..., a_{n-1}$*
3. *$p^2 \nmid a_0$*

Then $q(x)$ is irreducible in $\mathbb{Q}[x]$.

Example 1.1.13.

1. The polynomial
$$q(x) = 2x^3 + 6x^2 - 9x + 15$$
satisfies Eisenstein's criterion with $p = 3$.

2. The polynomial
$$q(x) = x^2 + x + 1$$
does not satisfy Eisenstein's criterion. Nevertheless, one can consider the transformation
$$r(x) = q(x+1) = (x+1)^2 + (x+1) + 1 = x^2 + 3x + 3,$$
which does satisfy the criterion. Hence $q(x)$ is also irreducible.

   More generally, using the same method as above, one can show that for any prime $p \in \mathbb{P}$, the polynomial
$$q(x) = x^{p-1} + x^{p-2} + ... + 1$$
is irreducible.

Example 1.1.14. We recall the following basic homomorphisms: 10.09.2026

1. Suppose R is a ring. Then there exists a unique ring homomorphism
$$\varepsilon : \mathbb{Z} \to \mathbb{R},\ n \mapsto n \cdot 1_R.$$

2. Suppose $\mathbb{F}$ is a field. Let $\varepsilon : \mathbb{Z} \to \mathbb{F}$ be the ring homomorphism from the above, and let $I = \ker \varepsilon$. Then ε induces a monomorphism $\mathbb{Z}/I \to \mathbb{F}$. In particular, $\mathbb{Z}/I$ is an integral domain, therefore $I \subseteq \mathbb{Z}$ is a prime ideal. Thus $I = \{0\}$ or $I = (p)$, for some $p \in \mathbb{P}$. This "formalizes" our previous definition of $\operatorname{char} \mathbb{F}$.
   i. If $\operatorname{char} \mathbb{F} = 0$, then $\varepsilon : \mathbb{Z} \to \mathbb{F}$ is a monomorphism. In particular, ε extends to a monomorphism $\mathbb{Q} \to \mathbb{F}$.
   ii. If $\operatorname{char} \mathbb{F} = p \in \mathbb{P}$, then $\operatorname{ran} \varepsilon \cong \mathbb{Z}/p\mathbb{Z} \cong \mathbb{F}_p$.

3. Suppose we have a fixed homomorphism of rings $\iota : R \to S$. Then for every choice of $\theta \in S$ there exists a unique ring homomorphism
$$\mathrm{ev}_\theta : R[x] \to S$$
such that $\mathrm{ev}_\theta(r) = \iota(r)$, for all $r \in R$, and $\mathrm{ev}_\theta(x) = \theta$, namely the evaluation map $p(x) \mapsto p(\theta)$.

Definition 1.1.15. Let $\mathbb{F}$ be a field. A *(field) extension* of $\mathbb{F}$ is a field $\mathbb{K}$ containing[2] $\mathbb{F}$ as a subfield. We will also write $\mathbb{F} \hookrightarrow \mathbb{K}$ to denote this relationship.

Example 1.1.16.

1. $\mathbb{Q} \hookrightarrow \mathbb{R}$
2. $\mathbb{R} \hookrightarrow \mathbb{C}$

If $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension, then $\mathbb{K}$ can be viewed as a vector space over $\mathbb{F}$.

Definition 1.1.17. The *degree* of a field extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is defined as the dimension of $\mathbb{K}$ as a vector space over $\mathbb{F}$. We also denote this as $[\mathbb{K} : \mathbb{F}]$ or $\dim_{\mathbb{F}} \mathbb{K}$.

Example 1.1.18.

1. $[\mathbb{R} : \mathbb{Q}] = \infty$
2. $[\mathbb{C} : \mathbb{R}] = 2$

Lemma 1.1.19. *Suppose we have field extensions $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$, then*

$$[\mathbb{L} : \mathbb{F}] = [\mathbb{L} : \mathbb{K}] \cdot [\mathbb{K} : \mathbb{F}].$$

Definition 1.1.20. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension and $\alpha \in \mathbb{K}$. We say that α is *algebraic* over $\mathbb{F}$ if there exists a non-zero polynomial $p(x) \in \mathbb{F}[x]$ such that $p(\alpha) = 0$. Otherwise α is called *transcendental* over $\mathbb{F}$.

Example 1.1.21.

1. $\sqrt{2}, \sqrt{3}$ are algebraic over $\mathbb{Q}$.
2. π, e are transcendental over $\mathbb{Q}$.

Definition 1.1.22. We say that an extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is *algebraic*, if every element $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Otherwise, we say that the extension is *transcendental*.

1.2 Minimal polynomials

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension and $\alpha \in \mathbb{K}$ is an algebraic element over $\mathbb{F}$. Then

$$I = \{p(x) \in \mathbb{F}[x] : p(\alpha) = 0\}$$

is a nonzero ideal in $\mathbb{F}[x]$. Since $\mathbb{F}[x]$ is a principle ideal domain, there exists a unique monic polynomial $m_{\alpha,\mathbb{F}}(x) \in \mathbb{F}[x]$, such that $I = \big(m_{\alpha,\mathbb{F}}(x)\big)$.

[2] Again, up to a monomorphism.

Definition 1.2.1. The polynomial $m_{\alpha,\mathbb{F}}(x)$ in the above is called the *minimal polynomial* of α over $\mathbb{F}$. If $\mathbb{F}$ is clear from the context, we also abbreviate this as $m_\alpha(x)$.

Lemma 1.2.2. *A monic polynomial $p(x) \in \mathbb{F}[x]$ is a minimal polynomial of $\alpha \in \mathbb{K}$ if and only if*

1. *$p(\alpha) = 0$, and*
2. *$p(x)$ is irreducible over $\mathbb{F}$.*

Proof.

"$\Longrightarrow$": Suppose $p(x) = m_\alpha(x)$. By definition, $p(\alpha) = 0$ and $p(x)$ generates the ideal of polynomials vanishing at α. Suppose $p(x) = q_1(x) \cdot q_2(x)$. Then $p(\alpha) = q_1(\alpha) \cdot q_2(\alpha)$. Without loss of generality, assume that $q_1(\alpha) = 0$, that is $q_1(x) \in (p(x))$, therefore $p(x) \mid q_1(x)$. But also $q_1(x) \mid p(x)$ by our assumption. Therefore $p(x)$ and $q_1(x)$ are associate in $\mathbb{F}[x]$, hence $q_2(x)$ is a unit.

"$\Longleftarrow$": Similarly to the above. ∎

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Since $m_{\alpha,\mathbb{F}}(x)$ is irreducible, the ideal $\big(m_{\alpha,\mathbb{F}}(x)\big)$ is maximal. Hence $\mathbb{F}[x]/\big(m_{\alpha,\mathbb{F}}(x)\big)$ is a field.

Proposition 1.2.3. *In the context above, $\mathbb{F}[x]/\big(m_{\alpha,\mathbb{F}}(x)\big) \cong \mathbb{F}(\alpha)$.*

Proof. Consider the evaluation homomorphism

$$\mathrm{ev}_\alpha : \mathbb{F}[x] \to \mathbb{K}.$$

The image of ev_α is $\mathbb{F}[\alpha]$, the subring of $\mathbb{K}$ generated by $\mathbb{F}$ and α. But the image is also isomorphic to $\mathbb{F}[x]/\big(m_{\alpha,\mathbb{F}}(x)\big) = \mathbb{F}[x]/\ker \mathrm{ev}_\alpha$, which is a field. ∎

Corollary 1.2.4. *In the context above, $[\mathbb{F}(\alpha) : \mathbb{F}] = \deg m_{\alpha,\mathbb{F}}$.*

Proof. This follows from the general fact that, for a monic polynomial $p(x) = a_0 + ... + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$, it holds that

$$\mathbb{F}[x]/(f(x)) \cong \left\{b_0 + b_1 x + ... + b_{n-1}x^{n-1} : b_0, ..., b_{n-1} \in \mathbb{F}\right\} \cong \mathbb{F}^n. \qquad ∎$$

Note that conversely, if $p(x) = a_0 + ... + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$ is an irreducible polynomial, then $\mathbb{F}[x]/\big(m_{\alpha,\mathbb{F}}(x)\big)$ is a field extension of $\mathbb{F}$ of degree n. Furthermore, the element

$x + (p(x)) \in \mathbb{F}[x]/(p(x))$ is a zero of $p(x)$, and $p(x)$ is the minimal polynomial of the element $x + (p(x))$.

Lemma 1.2.5. *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension, then $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if $[\mathbb{F}(\alpha) : \mathbb{F}] < \infty$.*

Proof.

"$\Longrightarrow$": Follows immediately from the previous corollary.

"$\Longleftarrow$": The elements $1, \alpha, \alpha^2, ..., \alpha^n$ are linearly dependant over $\mathbb{F}$. Therefore there exist $s_0, s_1, ..., s_n \in \mathbb{F}$, not all zero, such that $s_0 + s_1\alpha + ... + s_n\alpha^n = 0$, therefore α is algebraic over $\mathbb{F}$. ■

Corollary 1.2.6. *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension. If $[\mathbb{K} : \mathbb{F}] < \infty$, then $\mathbb{K}$ is algebraic over $\mathbb{F}$.*

Corollary 1.2.7. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\alpha, \beta \in \mathbb{K}$ are algebraic over $\mathbb{F}$. Then $\alpha \pm \beta, \alpha\beta, \frac{1}{\alpha}$ (provided that $\alpha \neq 0$) are algebraic over $\mathbb{F}$.*

Proof. By the previous lemma, it suffices to show that the respective generated extensions are of finite degree. The fields $\mathbb{F}(\alpha \pm \beta), \mathbb{F}(\alpha\beta)$ are subfields of $\mathbb{F}(\alpha, \beta)$. But the latter extension has finite degree, since

$$[\mathbb{F}(\alpha, \beta) : \mathbb{F}] = [\mathbb{F}(\alpha, \beta) : \mathbb{F}(\alpha)] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] \leq [\mathbb{F}(\beta) : \mathbb{F}] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] < \infty.$$

The final claim follows from the fact that $\mathbb{F}(\alpha) = \mathbb{F}(1/a)$. ■

Corollary 1.2.8. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$. Then the set of all algebraic elements of $\mathbb{F}$ is a sub-extension of $\mathbb{F}$, in other words it is a sub-field of $\mathbb{K}$ that is also an extension of $\mathbb{F}$. This is called the* algebraic closure *of $\mathbb{F}$ in $\mathbb{K}$.*

Example 1.2.9. We want to determine the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$. We consider that we ultimately want something like $x = \sqrt{2} + \sqrt{3}$. Squaring yields $x^2 + 5 + 2\sqrt{6}$, or equivalently $x^2 - 5 = 2\sqrt{6}$. Squaring again yields $x^4 - 10x^2 + 1$. It remains to show that this polynomail, say $p(x)$, is minimal, which will be the case if we can show that it is irreducible over $\mathbb{Q}$.

Since $p(x)$ is monic with integer coefficients, by Gauss' lemma it suffices to rule out factorizations over $\mathbb{Z}$. Note that, by the rational root theorem, the only root candidates are ± 1, and neither turn out to actually be roots. Hence there can be no linear factors.

Suppose that

$$\begin{aligned} p(x) &= x^4 - 10x^2 + 1 \\ &\stackrel{?}{=} (x^2 + ax + b)(x^2 + cx + d) \\ &= x^4 + ax^3 + bx^2 + cx^3 + acx^2 + bcx + dx^2 + adx + bd \\ &= x^4 + (a+c)x^3 + (b + ac + d)x^2 + (ad + bc)x + bd \end{aligned}$$

with $a, b, c, d \in \mathbb{Z}$. By comparing coefficients, we get

(i) $a + c = 0$,

(ii) $b + ac + d = -10$,

(iii) $ad + bc = 0$,

(iv) $bd = 1$.

From (i) we get $a = -c$, from (iii) we get $a(d - b) = 0$.

- If $a = 0$, then $c = 0$ and from (ii) we get $d = -10 - b$. From (iv) we get

$$1 \stackrel{?}{=} b(-10 - b) = -10b - b^2 \quad \Longleftrightarrow \quad b^2 + 10b - 1 = 0,$$

which yields $b = -5 \pm \sqrt{96} = -5 \pm 2\sqrt{6} \notin \mathbb{Q}$.

- If $d = b$, then from (iv) we get $1 = b^2$, thus $b = \pm 1$, and from (ii) we get $a^2 - 2b = 10$. The case $b = 1$ yields $a^2 = 12$, while $b = 1$ yields $a^2 = 8$, both of which are not perfect squares.

Therefore we have shown that $p(x)$ is irreducible over $\mathbb{Q}$, and hence the minimal polynomial of $\sqrt{2} + \sqrt{3}$.

Remark 1.2.10. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension and $\alpha_1, ..., \alpha_m \in \mathbb{K}$ are algebraic. Then $\mathbb{F}(\alpha_1, ..., \alpha_m)$ is a finite degree (and therefore algebraic) extension of $\mathbb{F}$.

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Lemma 1.2.11. *If $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\mathbb{K} \hookrightarrow \mathbb{L}$ are algebraic extensions, then the extension $\mathbb{F} \hookrightarrow \mathbb{L}$ is also algebraic.*

Proof. Let $\alpha \in \mathbb{L}$, then α is algebraic over $\mathbb{K}$. Therefore the exists a polynomial $p(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{K}[x]$ such that $p(\alpha) = 0$. Note that all $a_0, ..., a_n$ are algebraic over $\mathbb{F}$. Furthermore, $p(x) \in \mathbb{F}(a_0, ..., a_n)[x]$, so α is algebraic over $\mathbb{F}(a_0, ..., a_n)$. Therefore

$$[\mathbb{F}(a_0, ..., a_n, \alpha) : \mathbb{F}] = [\mathbb{F}(a_0, ..., a_n, \alpha) : \mathbb{F}(a_0, ..., a_n)] \cdot [\mathbb{F}(a_0, ..., a_n) : \mathbb{F}] < \infty. \qquad \blacksquare$$

1.3 Splitting fields

Definition 1.3.1. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension and $p(x) \in \mathbb{F}[x]$, $\deg p \geq 1$. We say that $p(x)$ splits completely over $\mathbb{K}$ if $p(x)$ is a product of linear polynomials over $\mathbb{K}$.

Example 1.3.2.

1. The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ splits completely over $\mathbb{R}$, since
$$x^2 - 2 = \left(x + \sqrt{2}\right)\left(x - \sqrt{2}\right).$$
2. The polynomial $x^2 + 2 \in \mathbb{Q}[x]$ splits completly over $\mathbb{C}$, since
$$x^2 + 2 = \left(x + i\sqrt{2}\right)\left(x - i\sqrt{2}\right).$$

Remark 1.3.3. If $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$ and $p(x) \in \mathbb{F}[x]$ splits completely over $\mathbb{K}$, then it also splits completely over $\mathbb{L}$.

Remark 1.3.4. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $p(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{F}[x]$. Then $p(x)$ splits completely over $\mathbb{K}$ if and only if there exist $\alpha_1, ..., \alpha_n \in \mathbb{K}$ such that

$$p(x) = a_n(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n).$$

Note that $\{\alpha_1, ..., \alpha_n\}$ is uniquely determined by $p(x)$.

Suppose $p(x) \in \mathbb{F}[x]$ splits completely over $\mathbb{K}$ as above. Consider the sub-extension $\mathbb{F} \hookrightarrow \mathbb{F}(\alpha_1, ..., \alpha_n) \hookrightarrow \mathbb{K}$. This intermediate extension has the following properties:

1. $p(x)$ splits completely over $\mathbb{F}(\alpha_1, ..., \alpha_n)$.
2. For any proper sub-extension $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{F}(\alpha_1, ..., \alpha_n)$, $p(x)$ does not split completely over $\mathbb{L}$.

Definition 1.3.5. Let $p(x) \in \mathbb{F}[x] \setminus \mathbb{F}$. A *splitting field* over $\mathbb{F}$ is any field extension $\mathbb{F} \hookrightarrow \mathbb{K}$ such that:

1. $p(x)$ splits completely over $\mathbb{K}$.
2. $p(x)$ does not split completely over any proper sub-extension $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{K}$.

We have already shown:

Lemma 1.3.6. *If $\mathbb{F} \hookrightarrow \mathbb{L}$ is an extension such that $p(x) \in \mathbb{F}[x] \setminus \mathbb{F}$ splits completely over $\mathbb{L}$, then $\mathbb{L}$ contains a splitting field of $p(x)$.*

Proposition 1.3.7. *For any $p(x) \in \mathbb{F}[x]$, $\deg p \geq 1$, there exists an extension $\mathbb{F} \hookrightarrow \mathbb{K}$ such that $p(x)$ splits completely over $\mathbb{K}$. It follows that every polynomial has a splitting field.*

Proof. Suppose $p(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{F}[x]$. We argue by induction on n.

- If $n = 1$, then $p(x)$ splits completely over $\mathbb{F}$.
- Suppose $n > 1$ and assume that any polynomial of degree $n-1$ over any field splits completely over some field extension. If $p(x)$ splits completely over $\mathbb{F}$ there is nothing to show. So assume that $p(x)$ does not split completely over $\mathbb{F}$. Then we can write (in $\mathbb{F}[x]$) that $p(x) = q(x) \cdot r(x)$, where $q(x)$ is irreducible over $\mathbb{F}$ and $\deg q > 1$. View $q(x)$ as a minimal polynomial of a root α and consider the induced extension $\mathbb{F} \hookrightarrow \mathbb{L}$. By assumption, we can write $q(x) = (x - \alpha) \cdot q_1(x)$. Consider the polynomial $q_1(x) \cdot r(x) \in \mathbb{L}[x]$, then $\deg(q_1 \cdot r) = n - 1$. By inducation hypothesis, there is an extension $\mathbb{L} \hookrightarrow \mathbb{K}$ such that $q_1(x) \cdot r(x)$ splits completely over $\mathbb{K}$. So we have extensions $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{K}$ and $p(x) = (x - \alpha) \cdot q_1(x) \cdot r(x)$. It follows that $p(x)$ splits completely over $\mathbb{K}$. ■

Remark 1.3.8. Suppose $\varphi : \mathbb{F} \to \mathbb{K}$ is a field isomorphism. Then φ induces a ring isomorphism

$$\varphi_x : \mathbb{F}[x] \to \mathbb{K}[x], p(x) \mapsto \varphi(p(x)).$$

A polynomial $p(x) \in \mathbb{F}[x]$ is irreducible if and only if $\varphi_x(p(x))$ is irreducible in $\mathbb{K}[x]$.

For any polynomial $p(x) \in \mathbb{F}[x]$, φ also induces an isomorphism of quotient rings

$$\mathbb{F}[x]/(p(x)) \cong \mathbb{K}[x]/(\varphi(p(x))).$$

Theorem 1.3.9. *Suppose $\varphi : \mathbb{K}_1 \to \mathbb{K}_2$ is an isomorphism of fields and $p(x) \in \mathbb{K}_1[x]$. Suppose $\mathbb{L}_1$ is any splitting field of $p(x)$ and $\mathbb{L}_2$ is any splitting field of $\varphi(p(x))$. Then there exists a field isomorphism $\psi : \mathbb{L}_1 \to \mathbb{L}_2$ such that the following diagram commutes:*

$$\begin{array}{ccc} & \overset{\psi}{\longrightarrow} & \\ \mathbb{L}_1 & & \mathbb{L}_2 \\ \uparrow & & \uparrow \\ \mathbb{K}_1 & \overset{\varphi}{\longrightarrow} & \mathbb{K}_2 \end{array}$$

In other words, φ can be extended to an isomorphism ψ of splitting fields.

Proof. Suppose $p(x) = a_0 + a_1x + ... + a_nx^n \in \mathbb{K}_1[x]$. We prove the statement by induction on n.

- If $n = 1$, then $\mathbb{L}_1 = \mathbb{K}_1$ and $\mathbb{L}_2 = \mathbb{K}_2$, thus there is nothing to show.
- Assume that the theorem holds if $\deg p < n$. If $p(x)$ splits completely over $\mathbb{K}_1$, there is nothing to prove. If we assume that $p(x)$ does not split completely over $\mathbb{K}_1$, then $\mathbb{K}_1 \subsetneq \mathbb{L}_1$, so we can choose some root $\alpha \in \mathbb{L}_1 \setminus \mathbb{K}_1$ of $p(x)$. We therefore have extensions

$$\mathbb{K}_1 \subsetneq \mathbb{K}_1(\alpha) \subseteq \mathbb{L}_1.$$

Let $m_{\alpha,\mathbb{K}_1}(x) \in \mathbb{K}_1[x]$ be the minimal polynomial of α. Then we have isomorphisms

$$\mathbb{K}_1[x]/\Big(m_{\alpha,\mathbb{K}_1}(x)\Big) \cong \mathbb{K}_1(\alpha),$$

and $\varphi_x\Big(m_{\alpha,\mathbb{K}_1}(x)\Big)$ is irreducible over $\mathbb{K}_2$. Therefore $\mathbb{K}_2[x]/\Big(\varphi_x\Big(m_{\alpha,\mathbb{K}_1}(x)\Big)\Big)$ is a field extension of $\mathbb{K}_2$. Note that $m_{\alpha,\mathbb{K}_1}(x) \mid p(x)$ and therefore $\varphi_x\Big(m_{\alpha,\mathbb{K}_1}(x)\Big) \mid \varphi_x(p(x))$. Since $\varphi_x(p(x))$ splits completly over $\mathbb{L}_2$, we have that $\varphi_x\Big(m_{\alpha,\mathbb{K}_1}(x)\Big)$ splits completely over $\mathbb{L}_2$. Therefore there is a field homomorphism $\mathbb{K}_2[x]/\Big(\varphi_x\Big(m_{\alpha,\mathbb{K}_1}(x)\Big)\Big)$ into $\mathbb{L}_2$, that sends x to a root of $\varphi_x\Big(m_{\alpha,\mathbb{K}_1}(x)\Big)$.

In $\mathbb{K}_1(\alpha)[x]$ we have $p(x) = (x-\alpha)\cdot q(x)$ for some $q(x) \in \mathbb{F}(\alpha)[x]$. Similarly, $\varphi_x(p(x)) = (x-\alpha_1)\cdot\varphi_x(q(x))$ in $\mathbb{K}_2(\alpha_1)[x]$. Since $p(x) = (x-\alpha)\cdot q(x)$ splits completely in $\mathbb{L}_1$, $q(x)$ must split completely in $\mathbb{L}_1$. On the other hand, $q(x)$ does not split completely in any proper subextension of $\mathbb{L}_1$, since if it did, then $p(x)$ would also split completely, contradicting the minimality of $\mathbb{L}_1$ for $p(x)$. Thus $\mathbb{L}_1$ is the splitting field of $q(x) \in \mathbb{K}_1(\alpha)[x]$. Similarly, $\mathbb{L}_2$ is the splitting field of $\varphi_x(q(x)) \in \mathbb{K}_2(\alpha_1)[x]$. But $\deg q < \deg p$, so by induction hypothesis there is an isomorphism $\psi : \mathbb{L}_1 \to \mathbb{L}_2$ extending $\mathbb{K}_1(\alpha) \to \mathbb{K}_2(\alpha_1)$, which in turn extends $\varphi : \mathbb{K}_1 \to \mathbb{K}_2$. ∎

Example 1.3.10.

1. The splitting field of $x^2 - 2 \in \mathbb{Q}[x]$ is $\mathbb{Q}\big(\sqrt{2}, -\sqrt{2}\big) = \mathbb{Q}\big(\sqrt{2}\big)$. Notice that $\big[\mathbb{Q}\big(\sqrt{2}\big) : \mathbb{Q}\big] = 2$.
2. Consider the polynomial $x^3 - 2 \in \mathbb{Q}[x]$. The field $\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}\big(\sqrt[3]{2}\big)$ contains a root of $x^3 - 2$. Therefore, in $\mathbb{Q}\big(\sqrt[3]{2}\big)[x]$, it holds that
$$x^3 - 2 = \big(x - \sqrt[3]{2}\big)\big(x^2 + \sqrt[3]{2}x + \sqrt[3]{4}\big).$$
The polynomial splits, but not completely. Indeed, the roots of $x^3 - 2$ are $\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2$, where $\omega = e^{\frac{2\pi i}{3}}$. The splitting field of $x^3 - 2$ therefore is $\mathbb{Q}\big(\sqrt[3]{2}, \omega\big)$. Note that, in this case, $\big[\mathbb{Q}\big(\sqrt[3]{2}, \omega\big) : \mathbb{Q}\big] = 3! \neq 3$.

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Lemma 1.3.11. *Suppose $\varphi : \mathbb{K}_1 \to \mathbb{K}_2$ is a field isomorphism and $\mathbb{F}_1 \hookrightarrow \mathbb{K}_1$ is a subfield. Let $\mathbb{F}_2 \coloneqq \varphi(\mathbb{F}_1)$. If $\mathbb{K}_1$ is a splitting field of some $p(x) \in \mathbb{F}_1[x] \setminus \mathbb{F}_1$, then $\mathbb{K}_2$ is a splitting field of $\varphi(p(x))$.*

Proof. Suppose $p(x) = a(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n)$ in $\mathbb{K}_1[x]$. Then
$$\varphi(p(x)) = \varphi(a)(x - \varphi(\alpha_1))(x - \varphi(\alpha_2))\cdots(x - \varphi(\alpha_n))$$
in $\mathbb{K}_2[x]$. So $\varphi(p(x))$ splits completely over $\mathbb{F}_2$.

If $\varphi(p(x))$ splits completly over $\mathbb{L}_2$ for some extension $\mathbb{F}_2 \hookrightarrow \mathbb{L}_2 \hookrightarrow \mathbb{K}_2$. Then $p(x)$ splits completely over $\mathbb{L}_1 \coloneqq \varphi^{-1}(\mathbb{L}_2)$. If $\mathbb{L}_2 \neq \mathbb{K}_2$, then $\mathbb{L}_1 \neq \mathbb{K}_1$, a contradiction. ∎

We prove a generalized converse:

Theorem 1.3.12. *Suppose we have:*

- *A field isomorphism $\varphi : \mathbb{F}_1 \to \mathbb{F}_2$.*
- *A polynomial $p(x) \in \mathbb{F}_1[x] \setminus \mathbb{F}_1$.*
- *A splitting field $\mathbb{K}_1$ of $p(x)$ over $\mathbb{F}_1$.*
- *A field extension $\mathbb{F}_2 \hookrightarrow \mathbb{L}_2$, where $\varphi(p(x))$ splits completely.*

Then there exists a field homomorphism $\psi : \mathbb{K}_1 \to \mathbb{L}_2$, extending φ. If $\mathbb{L}_2$ is a splitting field of $\varphi(p(x))$, then $\psi : \mathbb{K}_1 \to \mathbb{L}_1$ is an isomorphism.

Proof. To prove the last statement, assuming existence of ψ, note that $\varphi(\mathbb{K}_1) \hookrightarrow \mathbb{L}_2$ is a splitting field of $\varphi(p(x))$, by the previous lemma. So, if $\mathbb{L}_2$ is a splitting field, we have $\psi(\mathbb{K}_1) = \mathbb{L}_2$, so ψ is an isomorphism.

We prove the first statement by induction on $[\mathbb{K}_1 : \mathbb{F}_1]$.

- If $[\mathbb{K}_1 : \mathbb{F}_1] = 1$, then $\mathbb{K}_1 = \mathbb{F}_1$ and there is nothing to prove.
- Suppose $[\mathbb{K}_1 : \mathbb{F}_1] = n > 1$, and an appropriate ψ exists whenever the degree of an appropriate extension is less than n. We can write $p(x) = q(x) \cdot r(x)$ in $\mathbb{F}_1[x]$, where $q(x)$ is monic and irreducible with $\deg q > 1$. Since $p(x)$ splits completely in $\mathbb{K}_1[x]$, so $q(x)$ itself splits completely in $\mathbb{K}_1[x]$. Let $\alpha \in \mathbb{K}_1 \setminus \mathbb{F}_1$ be a root of $q(x)$. Similarly, $\varphi(q(x))$ is irreducible over $\mathbb{F}_2[x]$ and splits completely over $\mathbb{L}_1$, so let β be a root of $\varphi(q(x))$ in $\mathbb{L}_1$. The polynomials $q(x)$ and $\varphi(q(x))$ are minimal polynomials of α and β, respectively. We have a field isomorphisms

$$\mathbb{F}_1[x]/(q(x)) \to \mathbb{F}_1(\alpha), \quad \text{and} \quad \mathbb{F}_1[x]/(\varphi(q(x))) \to \mathbb{F}_2(\beta),$$

which, together with φ, induce a field isomorphism $\varphi' : \mathbb{F}_1(\alpha) \to \mathbb{F}_2(\beta)$. Note that $\mathbb{K}_1$ is a splitting field for $p(x)$ over $\mathbb{F}_1(\alpha)$. Moreover,

$$[\mathbb{K}_1 : \mathbb{F}_1(\alpha)] = \frac{[\mathbb{K}_1 : \mathbb{F}_1]}{[\mathbb{F}_1(\alpha) : \mathbb{F}_1]}) = \frac{[\mathbb{K}_1 : \mathbb{F}_1]}{\deg q} < [\mathbb{K}_1 : \mathbb{F}_1].$$

By the induction hypothesis, there exists a homomorphism $\psi : \mathbb{K}_1 \to \mathbb{L}_2$, extending φ' and therefore also φ. ■

Lemma 1.3.13. *Let $p(x) \in \mathbb{F}[x]$ with $\deg p = n$, and let $\mathbb{K}$ be a splitting field of $p(x)$. Then*

$$[\mathbb{K} : \mathbb{F}] \leq n!.$$

1.4 Splitting fields for a family of polynomials

Definition 1.4.1. Let $\mathbb{F}$ be a field, let $P = \{p_\alpha(x) : \alpha \in I\} \subset \mathbb{F}[x] \setminus \mathbb{F}$ be a family of polynomials. A splitting field for P is an extension $\mathbb{F} \hookrightarrow \mathbb{K}$, such that

1. Every polynomial $p_\alpha(x), \alpha \in P$ splits completely over $\mathbb{K}$.
2. $\mathbb{K}$ is minimal with respect to the first property.

Remark 1.4.2. If P only consists of finitely many, say $P = \{p_1(x), ..., p_n(x)\}$, then the splitting field for P is the same as the splitting field for $p_1(x) \cdots p_n(x)$.

Theorem 1.4.3. *For any family of polynomials, there exists a splitting field that is unique up to isomorphism of extensions.*

Proof. Let us prove existence when P is a countably infinite family, i.e. $P = \{p_1(x), p_2(x), ...\} \subset \mathbb{F}[x] \setminus \mathbb{F}$. Construct a sequence of extensions

$$\mathbb{F} = \mathbb{K}_0 \hookrightarrow \mathbb{K}_1 \hookrightarrow \mathbb{K}_2 \hookrightarrow \cdots \hookrightarrow \mathbb{K}_n \hookrightarrow \cdots,$$

where each $\mathbb{K}_n$ is the splitting for $p_n(x)$ over $\mathbb{K}_{n-1}$. Let

$$\mathbb{K} := \bigcup_{n=0}^{\infty} \mathbb{K}_n.$$

It is easily checked that $\mathbb{K}$ is indeed a field, and that each $p_n(x)$ splits completely over $\mathbb{K}$. We now claim that $\mathbb{K}$ is already the splitting field, which means it remains to show that it is minimal.

Suppose we have a proper sub-extension $\mathbb{F} \hookrightarrow \mathbb{L} \hookrightarrow \mathbb{K}$. Then there exists an $n \in \mathbb{N}$ such that $\mathbb{K}_{n-1} \subseteq \mathbb{L}$, but $\mathbb{K}_n \nsubseteq \mathbb{L}$. Then $\mathbb{L}$ is an extension of $\mathbb{K}_{n-1}$, but $\mathbb{L}$ does not contain a splitting field of $p_n(x)$ over $\mathbb{K}_{n-1}$ (since that splitting field is $\mathbb{K}_n$).

If P is uncountable, one can use Zorn's Lemma.

For uniqueness, suppose that $\mathbb{L}_1$ and $\mathbb{L}_2$ are splitting fields for P. Consider the set

$$X := \{(\mathbb{K}_1, \mathbb{K}_2, \varphi) : \mathbb{F} \hookrightarrow \mathbb{K}_1 \hookrightarrow \mathbb{L}_1, \mathbb{F} \hookrightarrow \mathbb{K}_2 \hookrightarrow \mathbb{L}_2, \varphi : \mathbb{K}_1 \to \mathbb{K}_2 \text{ is isomorphism}\}.$$

The set X can be partially ordered by considering extensions, and satisfies Zorn's Lemma. Therefore, there exists a maximal element $(\mathbb{K}_1^{\max}, \mathbb{K}_2^{\max}, \varphi^{\max})$. If $\mathbb{K}_1^{\max} \subsetneq \mathbb{L}_1$, then there exists a $p(x) \in P$ that does not split completely over $\mathbb{K}_1^{\max}$. Therefore we have splitting fields $\mathbb{K}_1^{\max} \subsetneq \mathbb{K}_1 \subseteq \mathbb{L}_1, \mathbb{K}_2^{\max} \subsetneq \mathbb{K}_2 \subseteq \mathbb{L}_2$ (for $p(x)$ and $\varphi^{\max}(p(x))$, respectively). So $\varphi^{\max}$ extends to an isomorphism $\mathbb{K}_1 \to \mathbb{K}_2$. ∎

Definition 1.4.4. Let $\mathbb{F}$ be a field. A splitting field of the family $\mathbb{F}[x] \setminus \mathbb{F}$ is called an *algebraic closure* of $\mathbb{F}$, and is often denoted $\overline{\mathbb{F}}$.

In particular, we now know that every field $\mathbb{F}$ has an algebraic closure, which is unique up to isomorphism.

Definition 1.4.5. A field $\mathbb{L}$ is *algebraically closed*, if every non-constant polynomial in $\mathbb{L}[x]$ splits completely over $\mathbb{L}$. Equivalently, every such polynomial has a root in $\mathbb{L}$.

Theorem 1.4.6. *The field $\mathbb{C}$ is algebraically closed.*

Theorem 1.4.7. *Suppose we have a field extension $\mathbb{F} \hookrightarrow \mathbb{K}$, then the following are equivalent:*

1. *$\mathbb{K}$ is an algebraic closure of $\mathbb{F}$.*
2. *Every non-constant polynomial over $\mathbb{F}$ splits completely over $\mathbb{K}$ and $\mathbb{K}$ an algebraic extension of $\mathbb{F}$.*
3. *$\mathbb{K}$ is algebraically closed and $\mathbb{K}$ is an algebraic extension of $\mathbb{F}$.*

1.5 Separability

01.10.2026

Let $\mathbb{F}$ be a field and $p(x) \in \mathbb{F}[x] \setminus \mathbb{F}$. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be an extension where $p(x)$ splits completely. Thus

$$p(x) = a(x-\alpha_1)\cdots(x-\alpha_n), \quad a \in \mathbb{F}^*, \quad \alpha_1, ..., \alpha_n \in \mathbb{K},$$

where the roots are not necessarily distinct.

Definition 1.5.1. A polynomial $p(x) \in \mathbb{F}[x]$ is called *separable* if in some (and therefore any) extension of $\mathbb{F}$ where $p(x)$ splits completely, the roots of $p(x)$ are pairwise distinct.

Example 1.5.2.

1. The polynomial $x^2 - 2x + 1 \in \mathbb{Q}[x]$ is not separable.
2. The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ factors as $\left(x - \sqrt{2}\right)\left(x + \sqrt{2}\right)$, and is therefore separable.

Does there exist an irreducible polynomial over $p(x) \in \mathbb{Q}[x]$ that is not separable? We will see later that this is not the case.

Definition 1.5.3. Let $p(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{F}[x]$. We define the *formal derivative* of $p(x)$ by

$$p'(x) := D(p(x)) := a_1 + 2a_2 x + ... + k a_k a^{k-1} + ... + n a_n x^{n-1}.$$

Lemma 1.5.4. *It holds that:*

- $D(p(x) + q(x)) = D(p(x)) + D(q(x))$
- $D(p(x)q(x)) = D(p(x))q(x) + D(q(x))p(x)$
- $D(p(q(x))) = p'(q(x))q'(x)$
- $\deg p'(x) \leq \deg p(x) - 1$

Lemma 1.5.5. *Let $p(x) \in \mathbb{F}[x]$ and $\mathbb{F} \hookrightarrow \mathbb{K}$. An element $\alpha \in \mathbb{K}$ is a multiple root of $p(x)$ in $\mathbb{K}[x]$ if and only if $(x-\alpha) \mid p(x)$ and $(x-\alpha) \mid D(p(x))$ in $\mathbb{K}[x]$.*

Proof. Suppose α is a multiple root. Then $p(x) = (x-\alpha)^2 q(x)$ for some polynomial $q(x) \in \mathbb{K}[x]$. But then

$$D(p(x)) = 2(x-\alpha)q(x) + (x-\alpha)^2 D(q(x)) = (x-\alpha)(2q(x) + (x-\alpha)D(q(x))),$$

hence $(x-\alpha) \mid D(p(x))$.

Conversely, suppose $(x-\alpha)$ divides both $p(x)$ and $D(p(x))$. Then we can write $p(x) = (x-\alpha)q(x)$, hence

$$D(p(x)) = q(x) + (x-\alpha)D(q(x)).$$

Since $(x-\alpha) \mid D(p(x))$ it also divides $q(x)$, therefore

$$p(x) = (x-\alpha)^2 r(x)$$

and α is a multiple root. ∎

Lemma 1.5.6. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $p(x), q(x) \in \mathbb{F}[x]$. Then*

$$\gcd_{\mathbb{F}[x]}(p(x), q(x)) = \gcd_{\mathbb{K}[x]}(p(x), q(x)).$$

Proof. Recall that a polynomial $d(x) \in \mathbb{F}[x]$ is a gcd of $p(x)$ and $q(x)$ if and only if (a) $d(x) \mid p(x)$ and $d(x) \mid q(x)$ in $\mathbb{F}[x]$ and (b) there exist $s(x), t(x) \in \mathbb{F}[x]$ such that $d(x) = s(x)p(x) + t(x)q(x)$. So let $d(x) = \gcd_{\mathbb{F}[x]}(p(x), q(x))$, then it is immediate that $d(x)$ satisfies (a) and (b) over $\mathbb{K}$ aswell. ∎

Proposition 1.5.7. *A polynomial $p(x) \in \mathbb{F}[x]$ is separable if and only if it is relatively prime (in $\mathbb{F}[x]$) to its derivative.*

Proof. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be an extension where $p(x)$ splits completely, so

$$p(x) = a(x-\alpha_1)\cdots(x-\alpha_n), \quad a \in \mathbb{F}^*, \quad \alpha_1, ..., \alpha_n \in \mathbb{K}.$$

By the previous lemma, it is enough to show that $\alpha_1, ..., \alpha_n$ are pairwise distinct if and only if $\gcd_{\mathbb{K}[x]}(p(x), D(p(x))) = 1$. If they are not distinct, then $p(x)$ has a multiple root α, and we have shown previously that $(x-\alpha) \mid \gcd_{\mathbb{K}[x]}(p(x), D(p(x)))$. If they are distinct, then an easy calculation shows that $\alpha_1, ..., \alpha_n$ are not roots of $D(p(x))$. So $p(x)$ and

$D(p(x))$ have no common roots. But since $p(x)$ splits completely in $\mathbb{K}[x]$, it follows that $\gcd_{\mathbb{K}[x]}(p(x), D(p(x))) = 1$. ∎

Corollary 1.5.8. *Suppose $p(x) \in \mathbb{F}[x]$ is irreducible. Then $p(x)$ is separable if and only if $D(p(x)) \neq 0$.*

Proof. If $p(x)$ is irreducible, then $\gcd(p(x), D(p(x)))$ is either 1 or $p(x)$. But $\deg D(p(x)) < \deg p(x)$, therefore $p(x)$ cannot be a divisor of $D(p(x))$, unless $D(p(x)) = 0$, in which case $\gcd(p(x), D(p(x))) = p(x)$. ∎

Corollary 1.5.9. *If* $\operatorname{char} \mathbb{F} = 0$*, then any irreducible polynomial over $\mathbb{F}$ is separable.*

Definition 1.5.10. A field $\mathbb{F}$ is called *perfect* if every irreducible polynomial over $\mathbb{F}$ is also separable.

Definition 1.5.11. An algebraic extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is called *separable* if for every $\alpha \in \mathbb{K}$, the minimal polynomial $m_{\alpha,\mathbb{K}}(x)$ is separable.

Example 1.5.12. Suppose $\mathbb{E}$ is a field of characteristic $p \in \mathbb{P}$, and $\alpha \in \mathbb{E}$ such that

$$\eta^p \neq \alpha, \quad \forall \eta \in \mathbb{E}.$$

Consider the polynomial $q(x) \coloneqq x^p - \alpha$. Then $D(q(x)) = px^{p-1} = 0$, so $q(x)$ is not separable. We can also verify this manually: let $\mathbb{E} \hookrightarrow \mathbb{L}$ be an extension where $q(x)$ splits completely. So there exists an element $\alpha^{\frac{1}{p}} \in \mathbb{L}$ such that $\left(\alpha^{\frac{1}{p}}\right)^p = \alpha \in \mathbb{E}$. In $\mathbb{L}[x]$, it holds that

$$x^p - \alpha = x^p - \left(\alpha^{\frac{1}{p}}\right)^p = \left(x - \alpha^{\frac{1}{p}}\right)^p.$$

We now want to show that $x^p - \alpha$ is irreducible in $\mathbb{E}[x]$. Suppose $x^p - \alpha = f(x)g(x)$ where $f(x) \in \mathbb{E}[x]$ is irreducible with $1 \leq \deg f(x) < p$. In $\mathbb{L}[x]$, the decomposition has the form

$$\left(x - \alpha^{\frac{1}{p}}\right)^p = \left(x - \alpha^{\frac{1}{p}}\right)^a \left(x - \alpha^{\frac{1}{p}}\right)^b,$$

where $1 \leq a < p$, therefore $\left(x - \alpha^{\frac{1}{p}}\right)^a \in \mathbb{E}[x]$. This shows that $\left(\alpha^{\frac{1}{p}}\right)^a \in \mathbb{E}$. Since $1 \leq a < p$ we can find $s, t \in \mathbb{Z}$ such that $sa + tp = 1$. Therefore

$$\alpha^{\frac{1}{p}} = \left(\alpha^{\frac{1}{p}}\right)^{sa+tp} \in \mathbb{E},$$

a contradiction to α not being a p-th power.

Example 1.5.13. Let $\mathbb{F}_p(t)$ be the quotient field of $\mathbb{F}_p[t]$. The elements of this field are expressions of the form

$$\frac{f(t)}{g(t)}, \quad f(t), g(t) \in \mathbb{F}_p[t], \quad g(t) \neq 0.$$

In this field, not every element is a p-th power. Indeed,

$$\left(\frac{a_0 + a_1 t + ... + a_n t^n}{b_0 + b_1 t + ... + b_m t^m}\right)^p = \frac{a_0^p + a_1^p t^p + ... + a_n t^{np}}{b_0^p + b_1^p t^p + ... + b_m t^{mp}}.$$

So, by the previous example, $x^p - t$ is an irreducible non-separable polynomial in $\mathbb{F}_p(t)[x]$.

Definition 1.5.14. Suppose char $\mathbb{F} = p$. Define the *Frobenius map* by

$$\sigma : \mathbb{F} \to \mathbb{F}, \alpha \mapsto \alpha^p.$$

Since the field has characteristic p, this map is a field monomorphism.

Theorem 1.5.15. *A field $\mathbb{F}$ of characteristic $p \in \mathbb{P}$ is perfect if and only the Frobenius map is an automorphism.*

Proof. We saw in the previous examples that, if σ is not surjective, then $\mathbb{F}$ is not perfect.

Conversely, suppose that σ is surjective, in particular an automorphism. Suppose towards a contradiction that $q(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{F}[x]$ is irreducible and non-separable. By a previous lemma, this implies that $D(q(x)) = 0$, therefore $\ell a_\ell = 0$ in $\mathbb{F}$ for all $\ell = 1, ..., n$. It follows that $a_k = 0$ whenever $p \nmid k$.

So $q(x) = a_0 + a_p x^p + a_{2p} x^{2p} + ... + a_{kp} x^{kp}$. Since σ is surjective, we can write $a_{\ell p} = b_\ell^p$ for some $b_\ell \in \mathbb{F}$, for $\ell = 0, ..., k$. Therefore

$$q(x) = b_0^p + (b_1 x)^p + \left(b_2 x^2\right)^p + ...\left(b_k x^k\right)^p = \left(b_0 + b_1 x + ... + b_k x^k\right)^p$$

and $q(x)$ is not irreducible, a contradiction. ∎

Corollary 1.5.16. *A finite field is perfect.*

Theorem 1.5.17. *For every $p \in \mathbb{P}$ and $k \in \mathbb{N}$ there exists a field, which we will denote $\mathbb{F}_{p^k}$ with p^k elements. Any two fields with p^k elements are isomorphic.*

Proof. Consider the polynomial $q(x) \coloneqq x^{p^k} - x \in \mathbb{F}_p[x]$ and satisfies $D(q(x)) = -1$, and is therefore separable. Let $\mathbb{F}_p \hookrightarrow \mathbb{K}$ be a splitting field of this polynomial. In $\mathbb{K}$, there are p^k distinct elements $\alpha_1, ..., \alpha_{p^k}$ that are roots of $q(x)$. We claim that these elements already constitute a subfield of $\mathbb{K}$.

- Clearly 0 and 1 are roots.
- If α, β are roots, then $(\alpha \pm \beta)^{p^k} = \alpha^{p^k} \pm \beta^{p^k} = \alpha \pm \beta$, as well as $(\alpha\beta)^{p^k} = \alpha\beta$ and $\left(\frac{\alpha}{\beta}\right)^{p^k} = \frac{\alpha}{\beta}$.

In this subfield $q(x)$ splits completely, and by minimality of $\mathbb{K}$ it follows that

$$\mathbb{K} = \left\{\alpha_1, ..., \alpha_{p^k}\right\}.$$

Therefore we can define $\mathbb{F}_{p^k}$ as the splitting field of $q(x)$ over $\mathbb{F}_p$.

On the other hand, if $\mathbb{E}$ is another field with p^k elements, then every $\alpha \in \mathbb{E}$ satisfies $\alpha^{p^k} = \alpha$, in other words, α is a root of $q(x)$. So $q(x)$ splits completely over $\mathbb{E}$, and since it cannot split in any smaller field (there are not enough elements) it follows that $\mathbb{E}$ is a splitting field of $q(x)$. ∎

1.6 Field homomorphisms (embeddings) and automorphisms

08.10.2026

Let $\mathbb{K}$ be a field and $\varphi : \mathbb{K} \to \mathbb{K}$ a field homomorphism[3]. Define the *field field* by

$$\mathbb{K}^\varphi \coloneqq \{x \in \mathbb{K} : \varphi(x) = x\}.$$

Lemma 1.6.1. *$\mathbb{K}^\varphi$ is a subfield of $\mathbb{K}$.*

Proof. Follows immediately by computation. ∎

Let $\mathrm{Aut}(\mathbb{K})$ be the group of automorphisms of $\mathbb{K}$ and suppose $H \subseteq \mathrm{Aut}(\mathbb{K})$. Define

$$\mathbb{K}^H \coloneqq \{x \in \mathbb{K} : \varphi(x) \in x \text{ for all } \varphi \in H\} = \bigcap_{\varphi \in H} \mathbb{K}^\varphi.$$

By the above lemma, it follows that $\mathbb{K}^H$ is a subfield of $\mathbb{K}$.

Example 1.6.2. Complex conjugation $c : \mathbb{C} \to \mathbb{C}, z \mapsto \overline{z}$ is a field automorphism, with $\mathbb{C}^c = \mathbb{R}$. Furthermore $\{\mathrm{id}, c\} \cong \mathbb{Z}_2$ is a group acting on $\mathbb{C}$ by automorphisms, and it holds that $\mathbb{C}^{\{\mathrm{id}, c\}} = \mathbb{R}$.

[3] Recall that φ is automatically a monomorphism, i.e. an embedding.

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension. Define

$$\mathrm{Aut}_{\mathbb{F}}(\mathbb{K}) \coloneqq \mathrm{Gal}_{\mathbb{F}}(\mathbb{K}) \coloneqq \mathrm{G}_{\mathbb{F}}(\mathbb{K}) \coloneqq \{\varphi \in \mathrm{Aut}(\mathbb{K}) : \varphi(x) = x \text{ for all } x \in \mathbb{F}\}.$$

Lemma 1.6.3. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$, $\varphi \in \mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$, $p(x) \in \mathbb{F}[x]$ and $\alpha \in \mathbb{K}$ is a root of $p(x)$. Then $\varphi(\alpha)$ is also of a root of $p(x)$.*

Proof. Suppose $p(x) = a_0 + a_1 x + ... + a_n x^n \in \mathbb{F}[x]$. Then

$$\begin{aligned} p(\varphi(\alpha)) &= a_0 + a_1\varphi(\alpha) + ... + a_n\varphi(\alpha)^n \\ &= \varphi(a_0) + \varphi(a_1)\varphi(\alpha) + ... + \varphi(a_n)\varphi(\alpha^n) \\ &= \varphi(p(\alpha)) = 0. \end{aligned}$$

■

Corollary 1.6.4. *The group $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$ acts by permutation on the set of roots of any polynomial over $\mathbb{F}$.*

Corollary 1.6.5. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is an algebraic extension. Then $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K}) = \mathrm{End}_{\mathbb{F}}(\mathbb{K})$.*

Proof. Let $\varphi : \mathbb{K} \to \mathbb{K}$ be a homomorphism fixing $\mathbb{F}$. We know that φ is injective, we only need to show surjectivity. Let $\alpha \in \mathbb{K}$, then φ permutes the set of roots of $m_{\alpha,\mathbb{F}}(x)$. Therefore there exists a root $\beta \in \mathbb{K}$ of $m_{\alpha,\mathbb{K}}(x)$ such that $\varphi(\beta) = \alpha$. ■

Remark 1.6.6. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension. We obtain a function Φ from the set of subgroups of $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$ to the set of intermediate fields between $\mathbb{F}$ and $\mathbb{K}$, via $\Phi(H) \coloneqq \mathbb{K}^H$ for a subgroup H of $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$. Conversely, we also obtain a function Γ in the other direction, by $\Gamma(\mathbb{E}) = \mathrm{Gal}_{\mathbb{E}}(\mathbb{K}) = \{\varphi \in \mathrm{Gal}_{\mathbb{F}}(\mathbb{K}) : \varphi(x) = x \text{ for all } x \in \mathbb{E}\}$ for an intermediate field $\mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{K}$.

However, in general Φ and Γ are not inverses.

Example 1.6.7. We consider the extension $\mathbb{Q} \hookrightarrow \mathbb{Q}\big(\sqrt[3]{2}\big) \hookrightarrow \mathbb{R}$. What is $\mathrm{Gal}_{\mathbb{Q}}\Big(\mathbb{Q}\big(\sqrt[3]{2}\big)\Big)$? More generally: Suppose $\mathbb{Q} \hookrightarrow \mathbb{L}$ is any extension, what is $\mathrm{Hom}_{\mathbb{Q}}\Big(\mathbb{Q}\big(\sqrt[3]{2}\big), \mathbb{L}\Big)$?

Answer: We have

$$\begin{aligned} \mathrm{Hom}_{\mathbb{Q}}\Big(\mathbb{Q}\big(\sqrt[3]{2}\big), \mathbb{L}\Big) &= \mathrm{Hom}_{\mathbb{Q}}(\mathbb{Q}[x]/(x^3 - 2), \mathbb{L}) \\ &= \{\beta \in \mathbb{L} : \beta^3 = 2\}. \end{aligned}$$

Therefore $\mathrm{Hom}_{\mathbb{Q}}\Big(\mathbb{Q}\big(\sqrt[3]{2}\big), \mathbb{R}\Big) = \{1\}$ and $\mathrm{Gal}_{\mathbb{Q}}\Big(\mathbb{Q}\big(\sqrt[3]{2}\big)\Big) = \{1\}$. However, $\Gamma(\mathbb{Q}) = \{1\}$ and $\Phi(\{1\}) = \Big(\mathbb{Q}\big(\sqrt[3]{2}\big)\Big)^{\{1\}} = \mathbb{Q}\big(\sqrt[3]{2}\big)$.

Remark 1.6.8. We have the following "formal" properties of Φ and Γ:

1. i. If $H_1 \subseteq H_2 \subseteq \mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$, then $\Phi(H_1) \supseteq \Phi(H_2)$.
   ii. If $\mathbb{F} \hookrightarrow \mathbb{E}_1 \hookrightarrow \mathbb{E}_2 \hookrightarrow \mathbb{K}$, then $\Gamma(\mathbb{E}_1) \supseteq \Gamma(\mathbb{E}_2)$.
2. i. For every $H \subseteq \mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$, it holds that $H \subseteq \Gamma(\Phi(H))$.
   ii. For every $\mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{K}$, it holds that $\mathbb{E} \hookrightarrow \Phi(\Gamma(\mathbb{E}))$.
3. i. For every element $\mathbb{E} \in \operatorname{im}\Phi$, it holds that $\Phi(\Gamma(\mathbb{E})) = \mathbb{E}$.
   ii. For every element $H \in \operatorname{im}\Gamma$, it holds that $\Gamma(\Phi(H)) = H$.

To the see the last property, suppose that $\mathbb{E} = \Phi(H)$ for some subgroup H. Then

$$\Phi(H) \subseteq \Phi(\Gamma(\Phi(H))),$$

but also $H \subseteq \Gamma(\Phi(H))$, therefore $\Phi(H) \supseteq \Phi(\Gamma(\Phi(H)))$. Therefore

$$\mathbb{E} = \Phi(H) = \Phi(\Gamma(\Phi(H))) = \Phi(\Gamma(E)).$$

So, while Γ and Φ are not inverses in general, they are bijections between their images.

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension. In general, by the above,

$$\mathbb{F} \subseteq \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})} \subseteq \mathbb{K}.$$

Theorem 1.6.9. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is an algebraic extension. Then the following are equivalent:*

1. $\mathbb{F} = \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})}$
2. *The extension is both normal and separable.*

Recall that an extension being normal means that it is the splitting field of a family of polynomials over $\mathbb{F}$, or equivalently, the minimal polynomial of every $\alpha \in \mathbb{K}$ splits completely over $\mathbb{K}$.

The extension being separable means that the minimal polynomial of every $\alpha \in \mathbb{K}$ splits into distinct linear factors over the splitting field.

Proof. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is normal and separable. Let $\alpha \in \mathbb{K} \setminus \mathbb{F}$. We need to show that there is a homomorphism $\varphi : \mathbb{K} \to \mathbb{K}$ fixing $\mathbb{F}$ pointwise but not α. Consider the minimal polynomial of $m_{\alpha,\mathbb{F}}(x) \in \mathbb{F}[x]$, which has degree ≥ 2. By assumption, $m_{\alpha,\mathbb{F}}(x)$ has another root $\beta \in \mathbb{K}$ with $\beta \neq \alpha$, then $m_{\alpha,\mathbb{F}}(x) = m_{\beta,\mathbb{F}}(x)$. It follows that the fields $\mathbb{F}(\alpha)$ and $\mathbb{F}(\beta)$ are isomorphic extensions of $\mathbb{F}$. Moreover, there is a (unique) isomorphism $\varphi_0 : \mathbb{F}(\alpha) \to$

$\mathbb{F}(\beta)$ which fixes $\mathbb{F}$ and satisfies $\varphi(\alpha) = \beta$. We obtain an embedding $\mathbb{F}(\alpha) \hookrightarrow \mathbb{F}(\beta) \subseteq \mathbb{K}$. Now, $\mathbb{K}$ is a splitting field over $\mathbb{F}(\alpha)$, therefore any embedding $\varphi_0 : \mathbb{F}(\alpha) \to \mathbb{K}$ fixing $\mathbb{F}$ extends to a homomorphism $\varphi : \mathbb{K} \to \mathbb{K}$ satisfying $\varphi(\alpha) = \beta \neq \alpha$.

Now suppose that $\mathbb{F} = \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})}$. Let $\alpha \in \mathbb{K}$ and $m_{\alpha,\mathbb{F}}(x)$ be the minimal polynomial of α. Let $\alpha_1, ..., \alpha_n$ denote the distinct roots of $m_{\alpha,\mathbb{F}}(x)$ and suppose $\alpha = \alpha_1$. The group $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$ acts on $\{\alpha_1, ..., \alpha_n\}$ by permutation. Consider the polynomial

$$p(x) = (x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n) \in \mathbb{K}[x].$$

For any $\varphi \in \mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$, it holds that $\varphi(p(x)) = p(x)$. Therefore

$$p(x) \in \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})}[x] = \mathbb{F}[x].$$

It follows that $m_{\alpha,\mathbb{F}}(x) \mid p(x)$. In fact $p(x) = m_{\alpha,\mathbb{F}}(x)$, because each α_k is a root of $m_{\alpha,\mathbb{F}}(x)$. In particular, $m_{\alpha,\mathbb{F}}(x)$ splits as a product of distinct linear factors. ∎

Definition 1.6.10. An algebraic extension is *Galois* if it satisfies any of the (equivalent) conditions of the theorem above.

Corollary 1.6.11. *If $\mathbb{F} \hookrightarrow \mathbb{K}$ is a Galois extension (i.e. algebraic, normal and separable), then Φ is injective and Γ is surjective.*

Proof. If $\mathbb{F} \hookrightarrow \mathbb{K}$ is normal and separable, then for every intermediate extension $\mathbb{E}$ it holds that also $\mathbb{K}$ is normal and separable over $\mathbb{E}$. So $\mathbb{E} = \mathbb{K}^{\mathrm{Gal}_{\mathbb{E}}(\mathbb{K})} = \Phi(\Gamma(\mathbb{E}))$. ∎

Theorem 1.6.12. *Let $\mathbb{K}$ be a field and let $G \subseteq \mathrm{Aut}(\mathbb{K})$ be a finite subgroup. Then*

$$\left[\mathbb{K} : \mathbb{K}^G\right] = |G|.$$

Corollary 1.6.13. *For any field extension $\mathbb{F} \hookrightarrow \mathbb{K}$, the finite subgroups of $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$ are in the image of Γ.*

Proof. Let G be a such a finite subgroup. We know that $G \subseteq \Gamma(\Phi(G)) = \mathrm{Gal}_{\mathbb{K}^G}(\mathbb{K})$. On the other hand

$$|\Gamma(\Phi(G))| = \left[\mathbb{K} : \mathbb{K}^{\Gamma(\Phi(G))}\right] = [\mathbb{K} : \Phi(\Gamma(\Phi(G)))] = [\mathbb{K} : \Phi(G)] = \left[\mathbb{K} : \mathbb{K}^G\right] = |G|,$$

and therefore $G = \Gamma(\Phi(G))$. ∎

Ultimately, if $\mathbb{K}/\mathbb{F}$ is a finite Galois extension, then Φ, Γ are inverse bijections.

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Proposition 1.6.14. *Let $\varphi_1, ..., \varphi_n : \mathbb{F} \to \mathbb{K}$ be pairwise distinct field homomorphisms. Suppose there exist $a_1, a_2, ..., a_n \in \mathbb{K}$ such that*

$$a_1\varphi_1(x) + a_2\varphi_2(x) + ... + a_n\varphi_n(x) = 0, \quad \text{for all } x \in \mathbb{F}.$$

Then $a_1 = a_2 = ... = a_n = 0$.

Proof. Suppose $a_1, ..., a_n$ is a non-trivial n-tuple that satisfies the above hypothesis. Suppose $a_{i_1}, ..., a_{i_k}$ is a non-trivial k-tuple that satisfies

$$a_{i_1}\varphi_{i_1}(x) + ... + a_{i_k}\varphi_{i_k}(x) = 0, \quad \text{for all } x \in \mathbb{F}.$$

Without loss of generality, we can assume k to be minimal. There exists an element $x_0 \in \mathbb{F}$ such that $\varphi_{i_1}(x_0) \neq \varphi_{i_k}(x_0)$. For all $x \in \mathbb{F}$ we have the identity

$$a_{i_1}\varphi_{i_1}(x_0)\varphi_{i_1}(x) + ... + a_{i_k}\varphi_{i_k}(x_0)\varphi_{i_k}(x) = 0,$$

$$a_{i_1}\varphi_{i_k}(x_0)\varphi_{i_0}(x) + ... + a_{i_k}\varphi_{i_k}(x_0)\varphi_{i_k}(x) = 0.$$

Subtracting yields

$$a_{i_1}\big(\varphi_{i_1}(x_0) - \varphi_{i_k}(x_0)\big)\varphi_{i_1}(x) + ... + a_{i_{k-1}}\big(\varphi_{i_{k-1}}(x_0) - \varphi_{i_k}(x_0)\big)\varphi_{i_{k-1}}(x) = 0, \quad \text{for all } x \in \mathbb{F},$$

contradicting minimality of k. ∎

Proof of (Theorem 1.88). Let $G = \{e, g_2, ..., g_n\}$ and $w_1, ..., w_m \in \mathbb{K}$. We want to prove:

1. If $m < n$ then $w_1, ..., w_m$ do not span $\mathbb{K}$ over $\mathbb{K}^G$.
2. If $m > n$ then $w_1, ..., w_m$ are linearly dependent over $\mathbb{K}^G$.

Suppose $m < n$. Consider the system of equations

$$(g_1w_1)x_1 + (g_2w_1)x_2 + ... + (g_nw_1)x_n = 0,$$

$$(g_1w_2)x_1 + (g_2w_2)x_2 + ... + (g_nw_2)x_n = 0,$$

$$\vdots$$

$$(g_1w_m)x_1 + (g_2w_m)x_2 + ... + (g_nw_m)x_n = 0.$$

This is a system of m equations with n unknowns. Since $m < n$ there exists a non-trivial solution $(\beta_1, ..., \beta_n) \in \mathbb{K}^n \setminus \{0\}$. So for all $a_1, a_2, ..., a_m \in \mathbb{K}^G$, the following holds

$$(g_1(a_1w_1))\beta_1 + ... + (g_n(a_1w_1))\beta_n = 0,$$

$$\vdots$$

$$(g_1(a_m w_m))\beta_1 + ... + (g_n(a_m w_m))\beta_n = 0.$$

It follows that

$$g_1\left(\sum_{i=1}^{m} a_i w_i\right)\beta_1 + ... + g_n\left(\sum_{i=1}^{m} a_i w_i\right)\beta_n = 0.$$

If $w_1, ..., w_m$ generate $\mathbb{K}$ over $\mathbb{K}^G$, it follows that $g_1, ..., g_n$ are linearly dependent over $\mathbb{K}$ as functions from $\mathbb{K}$ to $\mathbb{K}$, contradicting the previous proposition.

The second assertion is proved similarly. ∎

Theorem 1.6.15. *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a finite degree extension. The following are equivalent:*

1. *$\mathbb{K}$ is the splitting field of some separable polynomial over $\mathbb{F}$.*
2. *$\mathbb{F} = \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})}$*
3. *$[\mathbb{K} : \mathbb{F}] = |\operatorname{Gal}_{\mathbb{F}}(\mathbb{K})|$*

Proof. We already proved that (1) $\iff$ (2).

To prove that (2) $\implies$ (3), we just need to note that $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$ is finite.

To prove that (3) $\implies$ (2), we notice that

$$[\mathbb{K} : \mathbb{F}] = |\operatorname{Gal}_{\mathbb{F}}(\mathbb{K})| = \left[\mathbb{K} : \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})}\right].$$

Since $\mathbb{F} \subseteq \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})}$ it follows that $\left[\mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})} : \mathbb{F}\right] = 1$, so $\mathbb{F} = \mathbb{K}^{\mathrm{Gal}_{\mathbb{F}}(\mathbb{K})}$. ∎

Definition 1.6.16. We say that a finite extension is *Galois*, if it satisfies any of the (equivalent) conditions in the theorem above.

Theorem 1.6.17 (The Fundamental Theorem of Galois Theory)**.** *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a finite Galois extension. Then there order-reversing, mutally inverse bijections Φ, Γ of posets*

$$\{\text{Subgroups of } \mathrm{Gal}_{\mathbb{F}}(\mathbb{K})\} \longleftrightarrow \{\text{Intermediate fields } \mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{K}\}$$

with $\Phi(H) = \mathbb{K}^H$ and $\Gamma(\mathbb{E}) = \mathrm{Gal}_{\mathbb{E}}(\mathbb{K})$.

Proof. The only remaining to prove is that for all subgroups $H \subseteq \mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$ it holds that $\Gamma(\Phi(H)) = H$. In other words, $H = \mathrm{Gal}_{\mathbb{K}^H}(\mathbb{K}) =: \hat{H}$. In general $H \subseteq \hat{H}$. Also $\mathbb{K}^{\hat{H}} = \mathbb{K}^H$. It follows that $|H| = \left[\mathbb{K} : \mathbb{K}^H\right] = \left[\mathbb{K} : \mathbb{K}^{\hat{H}}\right] = |\hat{H}|$, therefore $H = \hat{H}$. ∎

Example 1.6.18 (Finite fields)**.** We know that $\mathbb{F}_{p^n}$ is the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$. The extension $\mathbb{F}_p \hookrightarrow \mathbb{F}_{p^n}$ is separable because $\mathbb{F}_p$ is perfect, so it is a Galois

extension. So $|\operatorname{Gal}_{\mathbb{F}_p}(\mathbb{F}_{p^n})| = [\mathbb{F}_{p^n} : \mathbb{F}_p] = n$. The Frobenius map $\Phi : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}, x \mapsto x^p$ is an element of the Galois group. Note that $\Phi^n = \mathrm{id}$. On the other hand, if $1 \leq k < n$, then Φ^k is not the identity. Indeed, if $\Phi^k = \mathrm{id}$ then $x^{p^k} - x = 0$ for every $x \in \mathbb{F}_{p^n}$. But this polynomial can have at most p^k roots. So Φ is a generator of the Galois group. It follows that

$$\operatorname{Gal}_{\mathbb{F}_p}(\mathbb{F}_{p^n}) \cong \mathbb{Z}/_n$$

is a cyclic group of order n.

It follows that there is a bijection of posets between subfields of $\mathbb{F}_{p^n}$ and subfields of $\mathbb{Z}/_n$. The latter is isomorphic to the set of positive divisors of n.

Suppose $k \mid n$ and $\mathbb{Z}/_k \subseteq \operatorname{Gal}_{\mathbb{F}_p}(\mathbb{F}_{p^n})$, then

$$\left[\mathbb{F}_{p^n} : \left(\mathbb{F}_{p^n}\right)^{\mathbb{Z}/_k}\right] = k.$$

It follows that $|\left(\mathbb{F}_{p^n}\right)^{\mathbb{Z}/_k}| = (p^n)^{\frac{1}{k}} = p^{\frac{n}{k}}$. So

$$\left(\mathbb{F}_{p^n}\right)^{\mathbb{Z}/_k} \cong \mathbb{F}_{p^{\frac{n}{k}}}.$$

Example 1.6.19. The poset of subfields of $\mathbb{F}_{16} = \mathbb{F}_{2^4}$ is isomorphic to the poset of subgroups of $\mathbb{Z}/_4$, which is isomorphic to the poset of divisors of 4.

Remark 1.6.20. Suppose $f(x) \in \mathbb{Z}[x]$ is of degree n. If the image of $f(x)$ in $\mathbb{F}_p[x]$ is irreducible of degree n for some p, then $f(x)$ is irreducible in $\mathbb{Z}[x]$ (and also in $\mathbb{Q}[x]$).

What about the converse? Suppose $f(x)$ is reducible of degree n over $\mathbb{F}_p$ for every prime p. Does it follow that $f(x)$ is reducible in $\mathbb{Z}[x]$? The answer is *no*.

Lemma 1.6.21. *The polynomial $x^4 + 1$ is reducible over $\mathbb{F}_p$ for every prime p, but is irreducible in $\mathbb{Q}[x]$ and $\mathbb{Z}[x]$.*

Proof. If $p = 2$, then $x^4 + 1 = (x + 1)^4$, so reducible. Suppose p is odd. We have

$$x^4 + 1 \mid x^8 - 1 \mid x^{p^2-1} - 1 \mid x^{p^2} - x$$

in $\mathbb{Z}[x]$ (or $\mathbb{F}_p[x]$). $\mathbb{F}_{p^2}$ is the splitting field of $x^{p^2} - x$. It follows that $x^4 + 1$ splits completely in $\mathbb{F}_{p^2}[x]$. So the splitting field of $x^4 + 1$ over $\mathbb{F}_p$ has degree at most 2. But

if x^4+1 is irreducible over $\mathbb{F}_p$, then its splitting field contains a copy of $\mathbb{F}_p[x]/(x^4+1)$, which has degree 4, a contradiction. ■

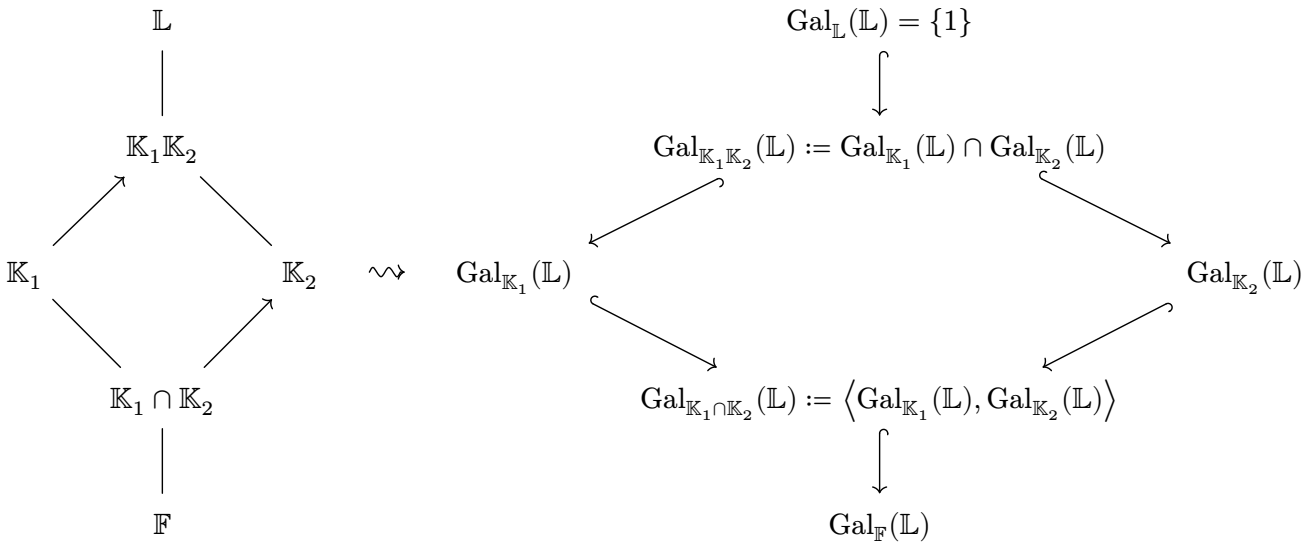
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Suppose $\mathbb{F} \hookrightarrow \mathbb{L}$ is a finite Galois extension and $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$ is an intermediate field. We note that:

1. $\mathbb{K}$ is always separable over $\mathbb{F}$ and $\mathbb{L}$ is separable over $\mathbb{K}$.
2. $\mathbb{L}$ is normal (and therefore Galois) over $\mathbb{K}$.
3. The following are equivalent:
   i. $\mathbb{K}$ is normal (Galois) over $\mathbb{F}$.
   ii. For every field homomorphism $f : \mathbb{K} \to \mathbb{L}$ fixing $\mathbb{F}$, it holds that $f(\mathbb{K}) \subseteq \mathbb{K}$.
   iii. $\mathrm{Gal}_{\mathbb{K}}(\mathbb{L})$ is a *normal* subgroup of $\mathrm{Gal}_{\mathbb{F}}(\mathbb{L})$.

In this case we have a homomorphism $\mathrm{Gal}_{\mathbb{F}}(\mathbb{L}) \to \mathrm{Gal}_{\mathbb{F}}(\mathbb{K})$ by restriction. It's kernel is $\mathrm{Gal}_{\mathbb{K}}(\mathbb{L})$, so we have $\mathrm{Gal}_{\mathbb{F}}(\mathbb{K}) \cong \mathrm{Gal}_{\mathbb{F}}(\mathbb{L})/\mathrm{Gal}_{\mathbb{K}}(\mathbb{L})$.

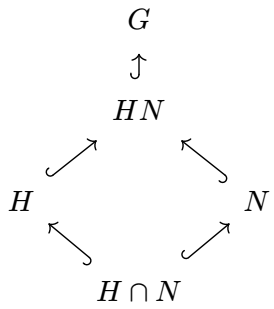
Now suppose we have two intermediate fields, $\mathbb{F} \hookrightarrow \mathbb{K}_1, \mathbb{K}_2, \hookrightarrow \mathbb{L}$. We can form a diagram of fields:



Recall from group theory: Suppose $H_1, H_2 \subseteq G$ are groups.

- $\langle H_1, H_2 \rangle \subseteq G$ is the subgroup generated by H_1 and H_2.
- $H_1 \cdot H_2 = \{h_1 \cdot h_2 : h_1 \in H_1, h_2 \in H_2\}$

If $H \subseteq G$ is any subgroup and $N \lhd G$ is a normal subgroup, then $HN = \langle H, N \rangle$. Therefore we obtain a diagram:

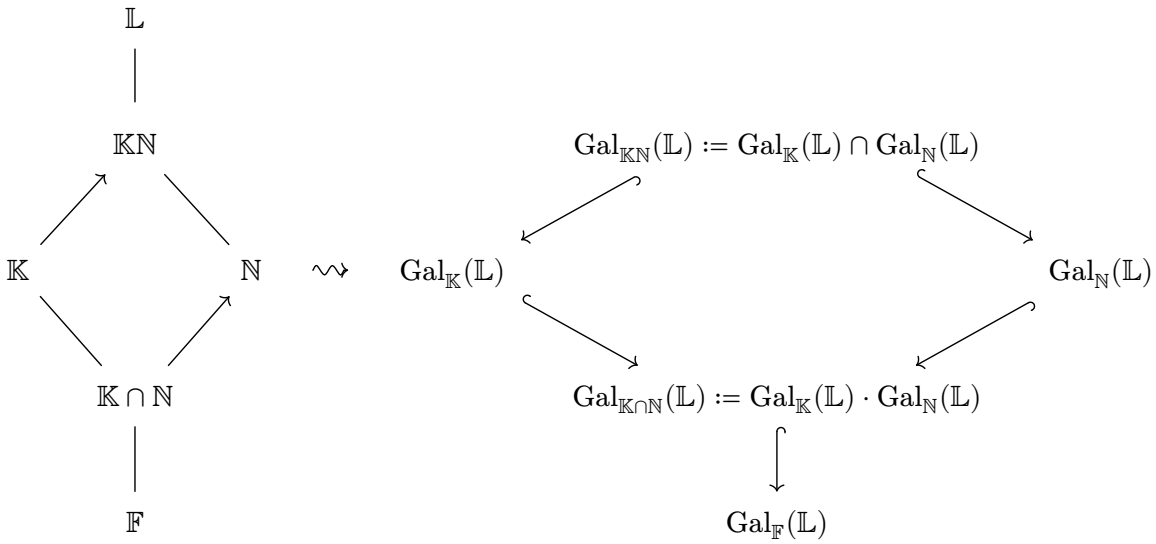


In this case $H \cap N$ is normal in H and there is an isomorphism $H/H \cap N \to HN/N$.

Now

Theorem 1.6.22. *Suppose* $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$ *is any field and* $\mathbb{F} \hookrightarrow \mathbb{N} \hookrightarrow \mathbb{L}$ *is Galois over* $\mathbb{F}$. *Then* $\mathbb{KN}$ *is Galois over* $\mathbb{K}$ *and* $\mathrm{Gal}_{\mathbb{K}}(\mathbb{KN}) \cong \mathrm{Gal}_{\mathbb{K}\cap\mathbb{N}}(\mathbb{N})$.

Proof. Consider the diagrams of fields and their Galois groups:



By the above, we observe that $\mathrm{Gal}_{\mathbb{KN}}(\mathbb{L})$ is normal in $\mathrm{Gal}_{\mathbb{K}}(\mathbb{L})$. Therefore $\mathbb{KN}$ is a normal extension of $\mathbb{K}$, and $\mathrm{Gal}_{\mathbb{K}}(\mathbb{KN}) = \mathrm{Gal}_{\mathbb{K}}(\mathbb{L})/\mathrm{Gal}_{\mathbb{KN}}(\mathbb{L})$. Therefore we have isomorphisms

$$\mathrm{Gal}_{\mathbb{K}}(\mathbb{KN}) \cong \mathrm{Gal}_{\mathbb{K}}(\mathbb{KN}) = \mathrm{Gal}_{\mathbb{K}}(\mathbb{L})/\mathrm{Gal}_{\mathbb{KN}}(\mathbb{L}) = \mathrm{Gal}_{\mathbb{KN}}(\mathbb{L})/\mathrm{Gal}_{\mathbb{N}}(\mathbb{L}) = \mathrm{Gal}_{\mathbb{K}\cap\mathbb{N}}(\mathbb{N}). \quad \blacksquare$$

Corollary 1.6.23. *In the set-up above*

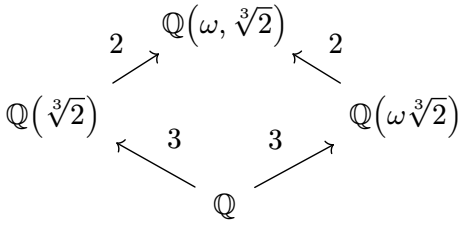
$$[\mathbb{KN} : \mathbb{F}] = \frac{[\mathbb{K} : \mathbb{F}] \cdot [\mathbb{N} : \mathbb{F}]}{[\mathbb{K} \cap \mathbb{N} : \mathbb{F}]}.$$

Proof. We compute

$$[\mathbb{KN} : \mathbb{F}] = [\mathbb{KN} : \mathbb{K}] \cdot [\mathbb{K} : \mathbb{K} \cap \mathbb{N}] \cdot [\mathbb{K} \cap \mathbb{N} : \mathbb{F}] = [\mathbb{N} : \mathbb{K} \cap \mathbb{N}] \cdot [\mathbb{K} : \mathbb{F}]$$
$$= \frac{[\mathbb{N} : \mathbb{F}] \cdot [\mathbb{K} : \mathbb{F}]}{[\mathbb{K} \cap \mathbb{N} : \mathbb{F}]}.$$
■

Example 1.6.24. We have $\mathbb{Q} \hookrightarrow \mathbb{Q}\left(\sqrt[3]{2}\right), \mathbb{Q}\left(\omega\sqrt[3]{2}\right) \hookrightarrow \mathbb{Q}\left(\omega, \sqrt[3]{2}\right)$, where the lastmost field is the splitting field of $x^3 - 2$, and $\omega \coloneqq e^{\frac{2\pi i}{3}}$.

We have a diagram:



Since $\mathbb{Q}\left(\sqrt[3]{2}\right), \mathbb{Q}\left(\omega\sqrt[3]{2}\right)$ are not normal over $\mathbb{Q}$, the formula does not apply.

1.7 Galois closures

Definition 1.7.1. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a finite separable extension. A *Galois closure* of $\mathbb{K}$ over $\mathbb{F}$ is a finite Galois extension $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$ such that no proper subextension $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{L}$ is Galois over $\mathbb{F}$.

Proposition 1.7.2. *Every finite separable extension $\mathbb{F} \hookrightarrow \mathbb{K}$ has a Galois closure $\mathbb{L}$, which is unique up to isomorphism fixing $\mathbb{F}$.*

Proof. Choose a finite set of generators $\alpha_1, ..., \alpha_n$ such that $\mathbb{K} = \mathbb{F}(\alpha_1, ..., \alpha_n)$. Let $m_1, ..., m_n$ be the minimal polynomials of $\alpha_1, ..., \alpha_n$ and let $\mathbb{L}$ be a splitting field of $\{p_1, ..., p_n\}$, or equivalently of $p_1 \cdot p_2 \cdot \cdots \cdot p_n$. Then $\mathbb{L}$ is separable and normal over $\mathbb{F}$, with $\alpha_1, ..., \alpha_n \in \mathbb{L}$. Therefore $\mathbb{K} = \mathbb{F}(\alpha_1, ..., \alpha_n) \hookrightarrow \mathbb{L}$.

If $\mathbb{F}$ is a proper subfield containing $\mathbb{K}$, then $p_1 \cdot \cdots \cdot p_n$ does not split completely over $\mathbb{E}$, but it has roots in $\mathbb{E}$, so $\mathbb{E}$ is normal. Moreover, $\mathbb{L}$ is a Galois closure of $\mathbb{K} = \mathbb{F}(\alpha_1, ..., \alpha_n)$ if and only if $\mathbb{L}$ is a splitting field of $p_1, ..., p_n$. So $\mathbb{L}$ is unique up to isomorphism. ■

Example 1.7.3. $\mathbb{Q}\left(\sqrt[3]{2}\right)$ is not Galois over $\mathbb{Q}$. The Galois closure is the splitting field of $x^3 - 2$, i.e. $\mathbb{Q}\left(\sqrt[3]{2}, \omega\right)$.

Remark 1.7.4. As an application, suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a finite separable extension. What are the intermediate fields? We can answer this using the above: Let $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$ be the Galois closure of $\mathbb{K}$. Then

$$\{\text{Intermediate fields } \mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{L}\} \longleftrightarrow \{\text{Subgroups of } \operatorname{Gal}_{\mathbb{F}}(\mathbb{K})\}.$$

It follows that the Galois correspondence restricts to a bijection

$$\{\text{Fields } \mathbb{E} \text{ s.t. } \mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{K}\} \longleftrightarrow \{\text{Intermediate groups } \operatorname{Gal}_{\mathbb{K}}(\mathbb{L}) \subseteq H \subseteq \operatorname{Gal}_{\mathbb{F}}(\mathbb{L})\}.$$

Example 1.7.5. What are the intermediate fields $\mathbb{Q} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{Q}\big(\sqrt[3]{2}\big)$? We conside the Galois closure $\mathbb{Q} \overset{3}{\hookrightarrow} \mathbb{Q}\big(\sqrt[3]{2}\big) \overset{2}{\hookrightarrow} \mathbb{Q}\big(\omega, \sqrt[3]{2}\big)$. Then

$$\operatorname{Gal}_{\mathbb{Q}} \mathbb{Q}\big(\omega, \sqrt[3]{2}\big) \cong S_3, \quad \operatorname{Gal}_{\mathbb{Q}\left(\sqrt[3]{2}\right)} \mathbb{Q}\big(\omega, \sqrt[3]{2}\big) \cong S_2.$$

The only intermediate groups $S_2 \subseteq H \subseteq S_3$ are S_2 and S_3, corresponding to the intermediate fields $\mathbb{Q}$ and $\mathbb{Q}\big(\sqrt[3]{2}\big)$.

1.8 Simple extensions

Definition 1.8.1. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a finite degree extension. We say that it is *simple* if $\mathbb{K} = \mathbb{F}(\alpha)$ for some $\alpha \in \mathbb{K}$. In other words, $\mathbb{K} = \mathbb{F}[x]/m_{\alpha,\mathbb{F}}(x)$.

Example 1.8.2.

1. $\mathbb{Q}\big(\sqrt{2}, \sqrt{3}\big) = \mathbb{Q}\big(\sqrt{2} + \sqrt{3}\big)$
2. $\mathbb{Q}\big(\sqrt[3]{2}, \omega\big) \overset{?}{\supseteq} \mathbb{Q}\big(\omega + \sqrt[3]{2}\big)$

Proposition 1.8.3. *A finite extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is simple if and only if there exist finitely many intermediate fields.*

Proof. Suppose $\mathbb{K} = \mathbb{F}(\alpha)$ is simple and let $m_{\alpha,\mathbb{F}}(x)$ be the minimal polynomial. Suppose $\mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{F}(\alpha)$, then $m_{\alpha,\mathbb{E}} \mid m_{\alpha,\mathbb{F}}$ in $\mathbb{E}[x]$ (and also in $\mathbb{K}[x]$). Suppose

$$m_{\alpha,\mathbb{E}}(x) = a_0 + a_1 x + ... + a_{k-1} x^{k-1} + x^k$$

for some $a_0, a_1, ..., a_{k-1} \in \mathbb{E}$. Let $\mathbb{E}' = \mathbb{F}(a_0, a_1, ..., a_{k-1})$. Then $\mathbb{E}' \subseteq \mathbb{E}$ and $m_{\alpha,\mathbb{E}'} = m_{\alpha,\mathbb{E}}$. We have

$$[\mathbb{K} : \mathbb{E}] = \deg m_{\alpha,\mathbb{E}} = \deg m_{\alpha,\mathbb{E}'} = [\mathbb{K} : \mathbb{E}']$$

and therefore $\mathbb{E} = \mathbb{E}'$. So every such $\mathbb{E}$ is $\mathbb{F}(a_0, ..., a_{k-1})$, where $a_0, ..., a_{k-1}$ are coefficients of some factor of $m_{\alpha,\mathbb{F}}$ in $\mathbb{K}[x]$. There are finitely many factors, so there are finitely many intermediate fields $\mathbb{E}$.

Conversely, suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ has finitely many intermediates, we want to prove $\mathbb{K} = \mathbb{F}(\alpha)$ for some $\alpha \in \mathbb{K}$. We differentiate between cases:

- **$\mathbb{K}$ is a finite field**. Then $\mathbb{K}^*$ is a cyclic group. Let $\alpha \in \mathbb{K}^*$ be a generator of $\mathbb{K}^*$ as a group, then $\mathbb{K} = \mathbb{F}(\alpha)$.
- **$\mathbb{F}, \mathbb{K}$ are infinite.** It suffices to show that $\mathbb{F}(\alpha, \beta) = \mathbb{F}(\gamma)$ for some $\gamma \in \mathbb{F}(\alpha, \beta)$. Consider all subfields of the form $\mathbb{F} \hookrightarrow \mathbb{F}(\alpha + c\beta) \hookrightarrow \mathbb{F}(\alpha, \beta)$, where $c \in \mathbb{F}$. It follows that there exist distinct $c_1, c_2 \in \mathbb{K}$ such that $\mathbb{F}(\alpha + c_1\beta) = \mathbb{F}(\alpha + c_2\beta)$. Then $(c_2 - c_1)\beta \in \mathbb{F}(\alpha + c_2\beta)$. Therefore $\beta \in \mathbb{F}(\alpha + c_2\beta)$ and $\alpha \in \mathbb{F}(\alpha + c_2\beta)$. Ultimately, $\mathbb{F}(\alpha, \beta) = \mathbb{F}(\alpha + c_2\beta)$. ■

Theorem 1.8.4. *A finite extension of a perfect field is simple.*

Proof. Suppose $\mathbb{F}$ is perfect, $\mathbb{F} \hookrightarrow \mathbb{K}$ is finite, then $\mathbb{K}$ is separable over $\mathbb{F}$. let $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$ be a Galois closure. Then intermediate fields $\mathbb{F} \hookrightarrow \mathbb{E} \hookrightarrow \mathbb{K}$ correspond to certain subgroups of $\mathrm{Gal}_{\mathbb{F}}(\mathbb{L})$, a finite group. ■

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Suppose $p(x) \in \mathbb{F}[x]$ is a separable polynomial. Consider the splitting field $\mathbb{E}$ of $p(x)$ over $\mathbb{F}$, then $\mathrm{Gal}_{\mathbb{F}}(\mathbb{E})$ permutes the roots of $p(x)$.

Question: Given a polynomial $p(x)$, what is the Galois group of $p(x)$, i.e. the Galois group of the splitting field? For example, suppose $u \in \mathbb{F}$. What is the Galois group of $x^n - u$?

In the following we consider the case $u = 1$, i.e. the Galois group of $x^n - 1$.

Solutions of the equation $x^n - 1 = 0$ in $\mathbb{F}$ are called *roots of unity.*

Remark 1.8.5. We observe the following:

1. In any field $\mathbb{F}$ there are at most n roots of unity.
2. If ξ, η are roots of unity, then so are $\xi\eta$ and $\frac{\xi}{\eta}$. Therefore, the set of roots of unity is a finite subgroup of $\mathbb{F}^*$. Such a subgroup is always cyclic.

3. Suppose $\mathbb{F}$ has d n-th roots of unity and ξ_d is a generator of this group. Then $\xi_d^d = 1$, but $\xi_d^k \neq 1$ for $1 \leq k < d$. Then d is the *order* of ξ_d in $\mathbb{F}^*$. But on the other hand, $\xi_d^n = 1$, therefore $d \mid n$.

Corollary 1.8.6. *The number of n-th roots of unity in any field $\mathbb{F}$ divides n.*

Definition 1.8.7. ξ_n is called a *primitive root of unity* if $\xi_n^n = 1$, but $\xi_n^k \neq 1$ for $1 \leq k < n$.

If $\mathbb{F}$ has a primitive root of unity ξ_n, then $1, \xi_n, \xi_n^2, ..., \xi_n^{n-1}$ are n distinct roots of unity. So every n-th root of unity is ξ_n^j for some j.

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\xi_n \in \mathbb{K}$ is a primitive root of unity. Then $\mathbb{F}(\xi_n)$ is a splitting field of $x^n - 1$ over $\mathbb{F}$.

We also note that if $\operatorname{char} \mathbb{F} \nmid n$ then the polynomial $x^n - 1$ is separable. Indeed, $D(x^n - 1) = nx^{n-1}$. If $\operatorname{char} \mathbb{F} \nmid n$ then $\gcd(x^n - 1, nx^{n-1}) = \gcd(x^n - 1, x^{n-1}) = 1$. It follows that $\mathbb{Q}(\xi_n)$ is a Galois extension of $\mathbb{Q}$. Such extensions are also called *cyclotomic extensions.*

ξ_n^j is a primitive root of unity if and only if $\gcd(j, n) = 1$. More generally:

Lemma 1.8.8. *ξ_n^j has order $\frac{n}{\gcd(j,n)}$ in $\mathbb{F}^*$.*

Note that

$$x^n - 1 = \prod_{\xi : \xi^n = 1} (x - \xi) = \prod_{j=0}^{n-1} (x - \xi_n^j)$$

in $\mathbb{Q}(\xi_n)[x]$.

Define

$$\Phi_n(x) = \prod_{\substack{\xi \text{ primitive } n\text{-th} \\ \text{root of unity}}} (x - \xi) = \prod_{\substack{1 \leq j \leq n-1 \\ \gcd(j,n)=1}} (x - \xi_n^j),$$

where in the lastmost expression ξ_n can be any fixed primitive n-th root.

Suppose $1 \leq d \mid n$. Then

$$\Phi_d(x) = \prod_{\substack{0 \leq j \leq d-1 \\ \gcd(j,d)=1}} \left(x - \xi_d^j\right) = \prod_{\substack{0 \leq j \leq d-1 \\ \gcd(j,d)=1}} \left(x - \xi_n^{\frac{nj}{d}}\right) = \prod_{\substack{0 \leq j \leq n-1 \\ \gcd(j,n)=d}} (x - \xi_n^j)$$

Corollary 1.8.9.

$$x^n - 1 = \prod_{j=0}^{n-1}(x - \xi_n^j) = \prod_{1\le d|n} \prod_{\substack{0\le j\le n-1 \\ \gcd(j,n)=d}} (x - \xi_n^j) = \prod_{1\le d|n} \Phi_d(x).$$

Example 1.8.10.

- $\Phi_1(x) = x - 1$
- $\Phi_2(x) = x + 1$
- $\Phi_3(x) = x - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} = x^2 + x + 1$
- $\Phi_4(x) = (x - i)(x + i) = x^2 + 1$

Proposition 1.8.11. *$\Phi_n(x)$ is a monic polynomial of degree $\varphi(n)$ with integer coefficients.*

Proof. We have

$$\Phi_n(x) = \prod_{\substack{0\le j\le n-1 \\ \gcd(j,n)=1}} (x - \xi_n^j),$$

so it is clearly monic of degree $\varphi(n)$.

Suppose by induction $\Phi_d(x)$ has integer coefficients for all $d < n$. Then

$$\Phi_n(x) = \frac{x^n - 1}{\prod_{\substack{d|n \\ 1\le d<n}} \Phi_d(x)},$$

where the denominator is a monic polynomial with integer coefficients. Since

$$\prod_{\substack{d|n \\ 1\le d<n}} \Phi_d(x) \mid x^n - 1 \quad \text{in } \mathbb{C}[x],$$

and $\prod_{1\le d|n} \Phi_d(x)$ is monic in $\mathbb{Z}[x]$, it follows that the quotient is in $\mathbb{Z}[x] \subset \mathbb{C}[x]$. Indeed since the denominator is monic, one can divide with remainder in $\mathbb{Z}[x]$. So

$$x^n - 1 = \prod_{\substack{d|n \\ 1\le d<n}} \Phi_{d(x)} q(x) + r(x)$$

where $q(x), r(x) \in \mathbb{Z}[x]$ and $\deg r < \deg \prod_{d|n, 1\le d<n} \Phi_d$. We know that $r(x) = 0$ in $\mathbb{C}[x]$, so $r(x) = 0$ in $\mathbb{Z}[x]$. So

$$q(x) = \frac{x^n - 1}{\prod_{\substack{d|n \\ 1 \le d < n}} \Phi_d(x)}$$

is in $\mathbb{Z}[x]$ and is equal to $\Phi_n(x)$. ∎

Theorem 1.8.12. *$\Phi_n(x)$ is irreducible in $\mathbb{Z}[x]$, and by Gauss' lemma therefore also in $\mathbb{Q}[x]$.*

Proof. Suppose towards a contradiction that $\Phi_n(x)$ is reducible, i.e. it can be written as $\Phi_n(x) = f(x)g(x)$ where $f(x), g(x)$ are monic in $\mathbb{Z}[x]$ and $f(x)$ is irreducible. Let ξ be a root of $f(x)$, then $\xi \in \mathbb{Q}(\xi_n) \subset \mathbb{C}$ and ξ is a primitive n-th root of unity. Let $p \in \mathbb{P}$ such that $p \nmid n$. Then ξ^p is another primitive n-th root of unity. Then

$$\Phi_n(\xi^p) = f(\xi^p)g(\xi^p) = 0.$$

We claim that $f(\xi^p) = 0$. Suppose towards a contradiction that $g(\xi^p) = 0$. Let $h(x) = g(x^p)$, then $h(\xi) = g(\xi^p) = 0$. But $f(x)$ is the minimal polynomial of ξ, therefore $f(x) \mid h(x)$, thus $f(x) \mid g(x^p)$. Let us reduce modulo p; let $\bar{f}(x), \bar{g}(x)$ etc. denote their images in $\mathbb{F}_p[x]$. Then $\bar{f}(x) \mid \bar{g}(x^p) = \bar{g}(x)^p$ and $\bar{f} = \bar{f}_1 \cdot \bar{f}_2$, where $\bar{f}_1 \in \mathbb{F}_p[x]$ is irreducible and $\bar{f}_1 \mid \bar{g}(x)$. Therefore we can write

$$\overline{\Phi_n}(x) = \bar{f}(x) \cdot \bar{g}(x) = \bar{f}_1(x)^2 \cdot \bar{h}(x),$$

where $\bar{f}_1(x) \in \mathbb{F}_{p[x]}$ is irreducible. Therefore $\overline{\Phi_n}(x)$ is not separable over $\mathbb{F}_p$. Moreover, $\overline{\Phi_n} \mid \overline{x^n - 1}$, so $\overline{x^n - 1}$ is not separable over $\mathbb{F}_p$. But $\gcd\left(\overline{x^n - 1}, \overline{nx^{n-1}}\right) = \gcd\left(\overline{x^n - 1}, \overline{x^{n-1}}\right) = 1$, so $\overline{x^n - 1}$ is separable, a contradiction. Therefore $f(\xi^p) = 0$.

So for every root ξ of $f(x)$ and every prime $p \nmid n$, it holds that ξ^p is another root of $f(x)$. Therefore, for any a such that $\gcd(a, n) = 1$, ξ^a is a root of $f(x)$. To see this, write $a = p_1 \cdots p_\ell$, where p_j are primes not dividing n. By iterating the argument, ξ^a is a root.

Therefore any primitive n-th root of unity is a root of $f(x)$. Therefore $f(x) = \Phi_n(x)$ and $\Phi_n(x)$ is reducible. ∎

Corollary 1.8.13.

1. *$\Phi_n(x)$ is the minimal polynomial of ξ_n.*
2. *There is an isomorphism of fields $\mathbb{Q}[x]/(\Phi_n(x))$ and $\mathbb{Q}(\xi_n)$ with $x \mapsto \xi_n$.*
3. *$[\mathbb{Q}(\xi_n) : \mathbb{Q}] = \deg \Phi_n(x) = \varphi(n)$*

Remark 1.8.14. For every a such that $\gcd(a,n)=1$, we have an automorphism

$$\begin{array}{ccc} \mathbb{Q}[x]/(\Phi_n(x)) & & \\ & \searrow^{x \mapsto \xi_n^a} & \\ \cong \big\downarrow & & \\ & \overset{\mu_a}{\searrow} & \\ \mathbb{Q}(\xi_n) & \longrightarrow & \mathbb{Q}(\xi_n) \end{array}$$

This satisfies

$$\mu_a \circ \mu_b(\xi_n) = \left(\xi_n^b\right)^a = \mu_{ab}(\xi_n).$$

The assignment $a \mapsto \mu_a$ defines a homomorphism

$$(\mathbb{Z}/_n)^* \to \mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\xi_n)).$$

This is a monomorphism, and since the sizes are the same it is an isomorphism. Therefore

$$\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\xi_n)) \cong (\mathbb{Z}/_n)^*.$$

Remark 1.8.15. If $n = p_1^{\ell_1} \cdots p_n^{\ell_n}$ where $p_1, ..., p_n$ are distinct primes, then

$$\mathbb{Z}/_n \cong \mathbb{Z}/_{p_1^{\ell_1}} \times \cdots \times \mathbb{Z}/_{p_n^{\ell_n}}$$

as rings, by the Chinese remainder theorem. Therefore

$$(\mathbb{Z}/_n)^* \cong \left(\mathbb{Z}/_{p_1^{\ell_1}}\right)^* \times \cdots \times \left(\mathbb{Z}/_{p_n^{\ell_n}}\right)^*.$$

By the primitive root theorem, if p is an odd prime, then

$$\left(\mathbb{Z}/_{p^\ell}\right)^* \cong \mathbb{Z}/_{p^\ell - p^{\ell-1}},$$

otherwise

$$(\mathbb{Z}/_{2^\ell})^* \cong \mathbb{Z}/_{2^{\ell-2}}.$$

From this, one can conclude:

Theorem 1.8.16. *For every finite abelian group A there exists an integer n and a subgroup $B \leq (\mathbb{Z}/_n)^*$ such that $(\mathbb{Z}/_n)^*/B \cong A$.*

Theorem 1.8.17. *For every finite abelian group A there exists a Galois extension $\mathbb{Q} \hookrightarrow \mathbb{F}$ with $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{F}) = A$. Namely, $\mathbb{F} = \mathbb{Q}(\xi_n)^B$, where B is as in the theorem above.*